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ON CRIMINAL ANTHROPOMETRY AND THE IDENTIFICATION OF CRIMINALS.

By W. R. MACDONELL, M.A., LL.D.

[Received November 5, 1901.]

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PART I.

MATERIAL AND METHODS.

(1) The object of this memoir is threefold :

(i) To test to what extent the criminal classes diverge in physical characters from other classes of the community.

(ii) To consider how far the shorter methods recently proposed by Professor Karl Pearson for finding the variability and correlation of characters in the case of normal frequency may be applied to some of the chief anthropometric measurements now customarily made, and

(iii) To determine what is the best manner in which these measurements can be applied to the identification of criminals.

I shall first consider the material I have had at my disposal ; I shall then indicate the methods I have used for the determination of its metrical constants, and finally apply my results to the consideration of the above three problems. In the course of my work I shall have to consider the important point of the homogeneity and normality of my material, and apply three separate tests :

(i) If broken up into groups, the statistical constants of a sub-group ought to remain, within the limits of random sampling, the same as the bulk of the observations.

(ii) Frequency distributions, if plotted and fitted with frequency curves, ought to give nearly normal distributions.

(iii) The regression lines in type cases ought to be closely represented by straight lines.

(2) *Nature of material used.* The data on which the memoir is based were obtained, through the kindness of Dr Garson, from the Central Metric Office, New Scotland Yard, where the register of habitual criminals is kept, and their identification effected. In an interesting paper published in the *Journal of the Anthropological Institute*, Vol. xxx. 1900, July—Dec., Dr Garson has explained in detail the metric system of identification in force in England, and the sort of prisoners whose metric description is registered at the Central Office, and whom we may call briefly “habitual” criminals.

But in addition to the official forms on which the descriptions of habitual criminals are recorded, the Central Office possesses a number of “practice” forms which record the metric description of a less pronounced type of criminal than the “habituals”—prisoners whose crimes and sentences are comparatively slight, and who may be called “non-habitual.” These practice forms are filled up by warders

on probation, and as their advancement in the Service depends on the accuracy with which they measure and describe the prisoners, they may be expected to do so with the utmost care. Indeed, I understand from Dr Garson that these "practice" forms may be accepted with as much confidence as the official forms for habitual criminals.

It is with these non-habitual criminals that we are concerned. Obviously, the larger the number of individuals we include, the better will be the results; on the other hand, the number must not be extremely large, otherwise the investigation becomes too laborious. Accordingly a group of 3000 was decided on as likely to give good results, while still keeping the work within manageable compass. Three thousand metric forms were thereupon obtained from Scotland Yard, containing particulars of 3000 male prisoners undergoing their sentences in the chief prisons of England and Wales. The majority of the prisoners were English and Welsh, many were Irish, and only a few Scotch; no foreigners or youths under 21 were included. The forms were drawn at random from the mass on the office shelves; we are therefore dealing with a random sampling.

The metric description recorded on the forms includes (*inter alia*) certain physical measurements, viz., those of the Head Length, Head Breadth, Face Breadth, Left Middle Finger, Left Cubit, Left Foot, and Height, of each individual. These measurements are taken, in the case of height, to the nearest $\frac{1}{8}$ of an inch, in the case of the other characters, to the nearest millimetre; thus, Head Length of 19.2 centimetres includes all head lengths from 19.15 cm. to 19.25 cm.; Height of 5 ft. 5 $\frac{1}{8}$ in. includes all heights from 5 ft. 5 $\frac{1}{16}$ in. to 5 ft. 5 $\frac{3}{16}$ in. These characters form the subject of the present memoir, and will be studied from the anthropometrical point of view in the earlier part of the paper; the problem of identification, with which Scotland Yard is concerned, will be discussed in the latter part.

(3) *Methods employed.* In dealing with this large mass of observations, I propose to apply the methods explained in Professor Karl Pearson's memoirs in the *Philosophical Transactions*, Vol. 195, A., pp. 1—47, 79—150, in order to calculate the Means, Standard Deviations, and Coefficients of Correlation, with their probable errors, for the above seven characters. These methods were devised by Professor Pearson to deal with characters not quantitatively measureable, but they are used here with the object of saving much of the labour involved in making these calculations in the ordinary way from the usual elaborate correlation tables. It will be seen later on that the saving of time and labour is very considerable.

To obtain the coefficients of correlation Professor Pearson shows (*loc. cit.* p. 2) that a 4-fold table is required; to find standard deviations and means, and probable errors and error correlations of all the quantities involved, a 9-fold table is required (*loc. cit.* pp. 82, 84). If a 16-fold table is formed, it will give the required 4-fold and 9-fold tables, as is shown in the following example, which is a frequency correlation table for the Head Length and Face Breadth of the 3000 criminals.

Head Length (in cms.)

Face Breadth (in cms.)	Class	18·8 and under	18·9 to 19·1	19·2 to 19·3	19·4 and over	Totals
	13·4 and under	453	222	135	229	1039
	13·5 to 13·6 ...	147	99	63	152	461
	13·7 to 13·8 ...	112	99	74	176	461
	13·9 and over	146	177	156	560	1039
	Totals	858	597	428	1117	3000

From this the following 9-fold table is constructed :

Head Length.

Face Breadth.	Class	18·8 and under	18·9 to 19·3	19·4 and over	Totals
	13·4 and under	453	357	229	1039
	13·5 to 13·8 ...	259	335	328	922
	13·9 and over	146	333	560	1039
	Totals	858	1025	1117	3000

and finally a 4-fold table to find coefficient of correlation :

Head Length.

Face Breadth.	Class	19·1 and under	19·2 and over	Totals
	13·6 and under	921	579	1500
	13·7 and over	534	966	1500
	Totals	1455	1545	3000

(It may be remarked that this latter table would also give the means, but we shall use the 9-fold table for this purpose.)

It seemed at the outset that the best results would be obtained by choosing the limits of divisions so that the totals of each column and of each row in the 4-fold and 9-fold tables should be as nearly as possible equal to a half and a third respectively of the whole 3000; but this principle was found in some cases to be inapplicable*.

(4) Before making use of these short tables we will first consider the results obtained in the case of our 3000 criminals from three frequency correlation tables (Tables I. II. and III.) made up in the usual way. The methods of preparing the tables and calculating the results, with their probable errors, are very fully explained

* In fixing the limits much help was obtained from Dr Garson's results.

in Mr G. Udny Yule's paper "On the Theory of Correlation" in the *Journal of the Royal Statistical Society*, Vol. LX., Part IV., December 1897. We have modified Mr Yule's formula for the square of the standard deviation by diminishing it by the quantity $\frac{1}{12}$, in accordance with Mr W. F. Sheppard's paper in the *Proceedings of the London Mathematical Society*, Vol. XXIX., Nos. 634/5.

The results are as follows :

TABLE 1.
3000 Criminals.

	Standard Deviation	Mean
Head Length (cm.) ...	$\cdot 6046 \pm \cdot 0053$	$19\cdot 1663 \pm \cdot 0075$
Head Breadth (cm.)...	$\cdot 5014 \pm \cdot 0044$	$15\cdot 0442 \pm \cdot 0062$
L. M. Finger (cm.) ...	$\cdot 5479 \pm \cdot 0048$	$11\cdot 5474 \pm \cdot 0068$
Height (ins.) ...	$2\cdot 5410 \pm \cdot 0221$	$65\cdot 5355 \pm \cdot 0313$

TABLE 2.
3000 Criminals.

	Coefficient of Correlation
Head Length and Head Breadth	$\cdot 4016 \pm \cdot 0103$
Head Breadth and Height ...	$\cdot 1831 \pm \cdot 0119$
L. M. Finger and Height ...	$\cdot 6608 \pm \cdot 0069$

(5) *Tests of Normality.* At the foot of Tables II. and III. (see pp. 215, 216) are shown the mean Finger Length and mean Head Breadth of each column array; these means are plotted on Figs. 1 and 2, and the lines are drawn which show the theoretical regression of Finger on Height, and Head Breadth on Height. The slope of these lines is calculated from the formula $\tan \theta = \frac{r\sigma_2}{\sigma_1}$, where r is the coefficient of correlation, and σ_1, σ_2 , the standard deviations of the correlated organs. (See Yule, *loc. cit.*) The means of the column arrays at the extremities of the tables are not included, as they are based on such small frequencies.

It will be observed that the regression lines fit the observations very well, i.e., the regression is very closely linear.

FIG. 1. Line of Regression: Left Middle Finger on Height. 3000 Criminals.
Mean Height 65·5. Mean Finger 115·5. $\tan \theta = 1·425$.

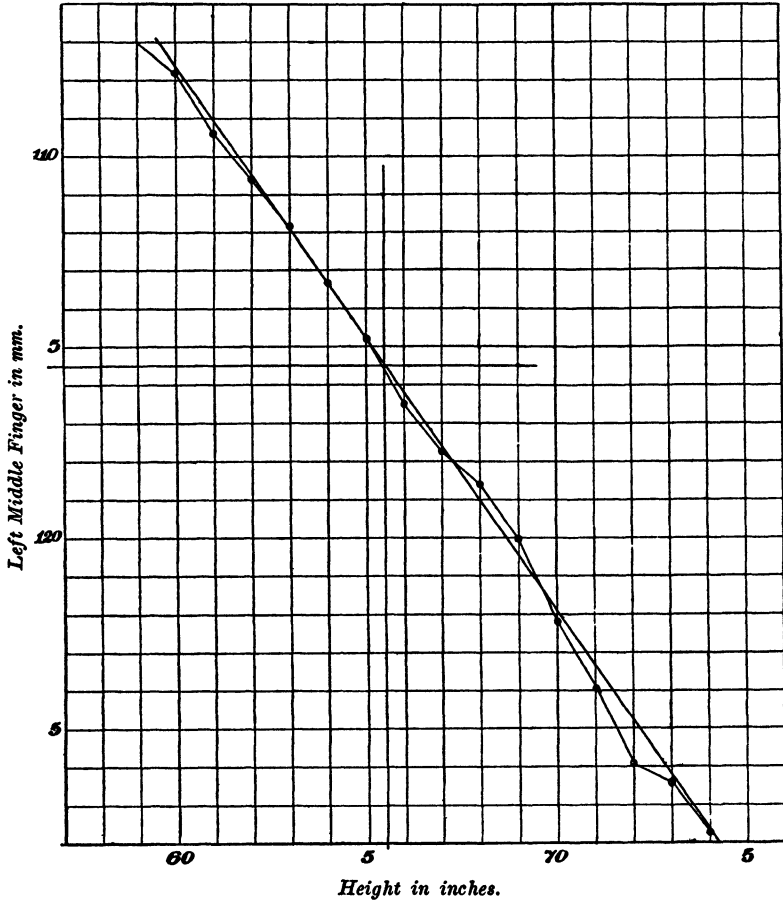
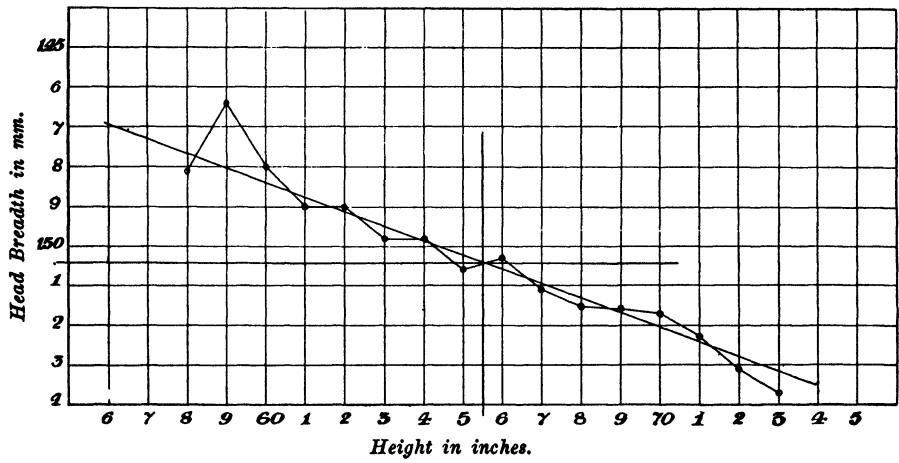


FIG. 2. Line of Regression: Head Breadth on Height. 3000 Criminals.
Mean Height 65·5. Mean Breadth 150·4. $\tan \theta = ·3612$.



We will next attempt to fit a curve to the observed frequencies of Head Breadth and Height in the case of the 3000 criminals in accordance with Professor Pearson's method (see *Phil. Trans.* Vol. 186, A. pp. 343—414).

(i) Head Breadth.

The moments are :

$$\begin{aligned}\mu_2 &= 25.14526 & \beta_1 &= .010385 \\ \mu_3 &= 12.84962 & \beta_2 &= 3.032609 \\ \mu_4 &= 1917.47151.\end{aligned}$$

Therefore the criterion $2\beta_2 - 3\beta_1 - 6 = +.03406$; the curve is therefore of Type IV., and the remaining constants are :

$$\begin{aligned}d &= - .253365 & \text{Skewness} &= - .050526 \\ r &= 356.203035 & m &= 179.101518 \\ \nu &= -194.9783 & a &= 82.900557 \\ \log y_0 &= \overline{20}3543484.\end{aligned}$$

The equation to the curve is :

$$\begin{aligned}x &= 82.900557 \tan \theta, \\ \log y &= \overline{20}3543484 + 358.203035 \log \cos \theta + 1.477904 \theta.\end{aligned}$$

The maximum ordinate or mode is 239.39 and occurs at $x = 45.124714$, corresponding to a head breadth of 15.0188; the mean or centroid occurs at $x = 45.378079$, its value being 15.0442.

The curve obviously approximates very closely to the normal curve.

(ii) Height.

$$\begin{aligned}\mu_2 &= 6.45660 & \beta_1 &= .002578 \\ \mu_3 &= .83300 & \beta_2 &= 3.175076 \\ \mu_4 &= 132.36172. \\ 2\beta_2 - 3\beta_1 - 6 &= +.342418.\end{aligned}$$

The curve is again of Type IV., and the remaining constants are :

$$\begin{aligned}d &= - .058071 & \text{Skewness} &= - .022853 \\ r &= 38.067473 & m &= 20.033736 \\ \nu &= -2.870793 & a &= 15.42649 \\ y_0 &= 433.647.\end{aligned}$$

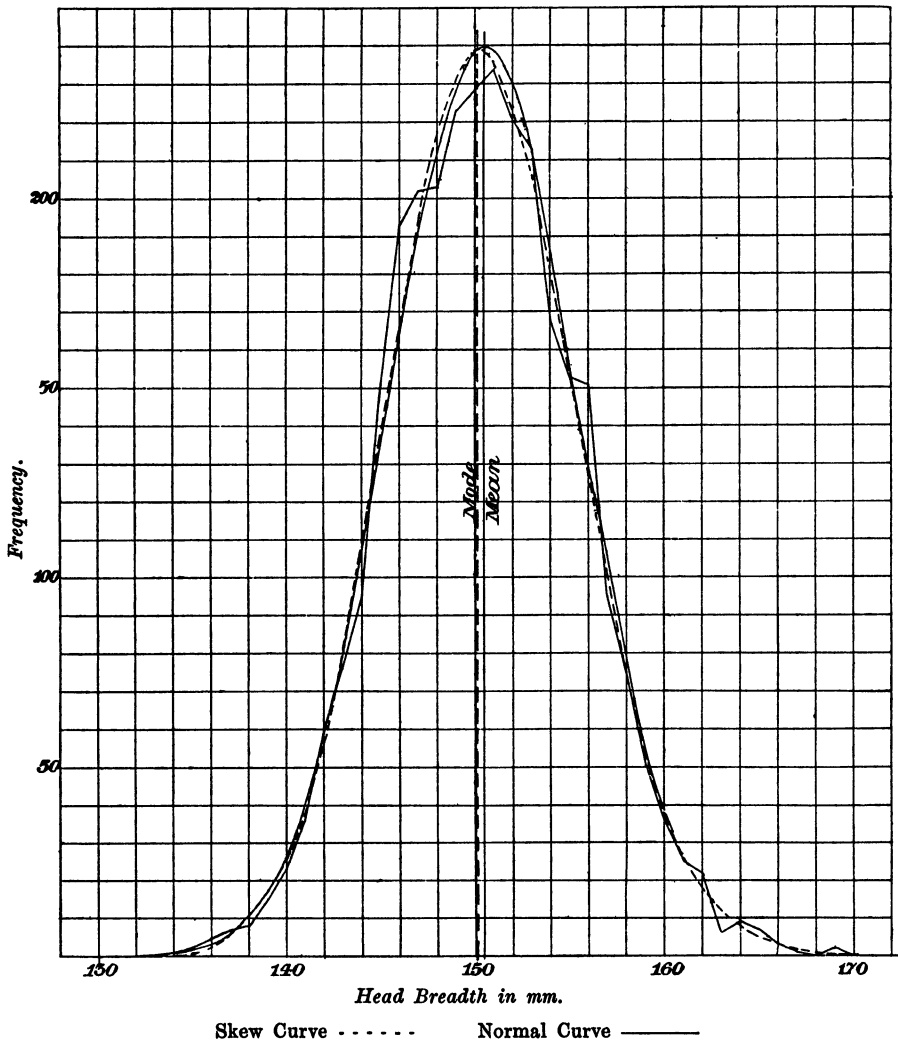
The equation to the curve is :

$$\begin{aligned}x &= 15.42649 \tan \theta, \\ y &= 433.647 (\cos \theta)^{40.067473} e^{2.870793 \theta}.\end{aligned}$$

The mode is 480.574 and occurs at $x = 1.105291$, corresponding to a height of 65''4774; the mean occurs at $x = 1.163362$, its value being 65''5355.

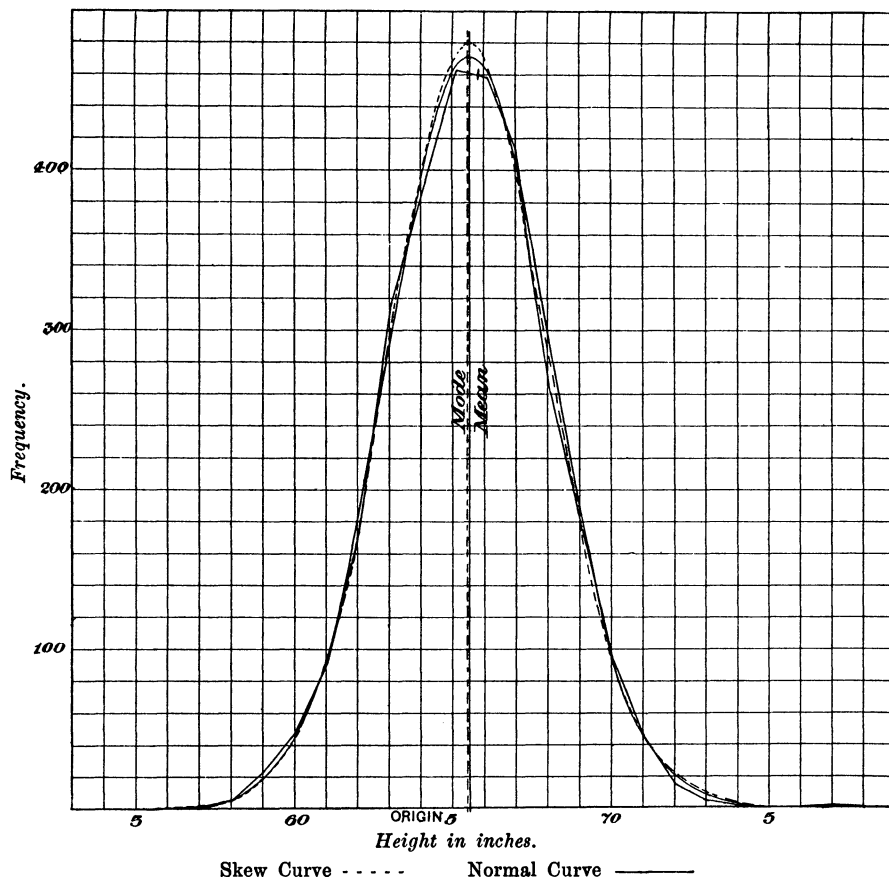
This curve also is very nearly normal. Both curves are drawn on Figs. 3 and 4, which bring out clearly the closeness of the two curves to each other.

FIG. 3. Head Breadth in Criminals.



Seeing that the regression lines are closely linear, and the frequency curves closely normal, we may assert that the distribution is at any rate very approximately normal. This is an important result, as it gives us confidence in applying Professor Pearson's abbreviated method, based as it is on the assumption of normal frequency.

FIG. 4. Height of Criminals.



(6) *Tests of Homogeneity.* In order to see whether or not we are dealing with a homogeneous class, three frequency correlation tables (Tables IV. V. and VI.; see pp. 217—219) were formed from the first batch of forms taken at random from Scotland Yard, and comprising particulars of 1306 out of our 3000 criminals. These tables give the following results, the method of calculation being the same as before:

TABLE 3.
1306 *Criminals.*

	Standard Deviation	Mean
Head Length (cm.) ...	$\cdot 6002 \pm \cdot 0079$	$19\cdot 1438 \pm \cdot 0112$
Head Breadth (cm.) ...	$\cdot 5133 \pm \cdot 0068$	$15\cdot 0302 \pm \cdot 0096$
L. M. Finger (cm.) ...	$\cdot 5451 \pm \cdot 0072$	$11\cdot 5244 \pm \cdot 0102$
Height (ins.) ...	$2\cdot 5906 \pm \cdot 0342$	$65\cdot 4193 \pm \cdot 0407$
Face Breadth (cm.) ...	$\cdot 5022 \pm \cdot 0066$	$13\cdot 6350 \pm \cdot 0094$

TABLE 4.

1306 *Criminals*.

	Coefficient of Correlation
Head Length and Head Breadth	$\cdot 3958 \pm \cdot 0157$
Head Breadth and Height ...	$\cdot 1973 \pm \cdot 0179$
L. M. Finger and Height ...	$\cdot 6682 \pm \cdot 0103$

To compare the results thus obtained for the two sets of 3000 and 1306, we will compare the absolute difference of their standard deviations, means, and coefficients of correlation with the square root of the sums of the squares of the probable errors. Let us call the difference D , the square root in question S , the coefficient of correlation r ; and to show the form of the calculation we will take the standard deviation of Head Length as an example. For the sake of keeping as many significant figures as possible, we will use the millimetre as our unit.

$$\begin{aligned}
 \text{s.D. of 1306 criminals is} & \quad 6\cdot00247 \pm \cdot 07922 \\
 \text{,, 3000 ,,} & \quad 6\cdot04593 \pm \cdot 05265 \\
 D = & \quad \cdot 04346 \\
 \{(\cdot 07922)^2 + (\cdot 05265)^2\}^{\frac{1}{2}} = S = & \quad \cdot 09512
 \end{aligned}$$

Thus it appears that D is less than one-half of S .

Collecting all our results together, we have the following tables:

TABLE 5.

1306 and 3000 *Criminals*.

	Standard Deviation		Mean	
	D	S	D	S
Head Length... ..	$\cdot 04346$	$\cdot 09512$	$\cdot 22535$	$\cdot 13451$
Head Breadth	$\cdot 11840$	$\cdot 08059$	$\cdot 13922$	$\cdot 11397$
Height... ..	$\cdot 04963$	$\cdot 04072$	$\cdot 11618$	$\cdot 05759$
L. M. Finger	$\cdot 02758$	$\cdot 08632$	$\cdot 22941$	$\cdot 12208$

TABLE 6.
1306 and 3000 *Criminals*.

	Coefficient of Correlation	
	<i>D</i>	<i>S</i>
Head Length and Head Breadth	·00586	·01884
Head Breadth and Height ...	·01420	·02154
Finger and Height ...	·00735	·01245

From these figures it is apparent that there is no sensible difference between the 1306 and the 3000; *D* is not in any case, *even in the means*, twice to thrice *S*. We may therefore treat our material for practical purposes as *normal* and *homogeneous*.

Before leaving this group of 1306, I may add that having formed correlation tables for Head Length and Left Middle Finger, Head Length and Face Breadth, I find the coefficients of correlation to be as follows:

TABLE 7.
1306 *Criminals*.

	Coefficient of Correlation
Head Length and Finger ...	·2861 ± ·0171
Head Length and Face Breadth	·4074 ± ·0156

(7) *Comparison of classes from which criminals are drawn with the educated classes of the community.* It will now be interesting to compare our 3000 criminals with an entirely different social class of the population, and for this purpose I have prepared frequency correlation Tables VII., VIII. and IX. showing the correlation between Head Length and Head Breadth, Height and Head Breadth, Height and Head Length, in the case of 1000 male students at Cambridge. The figures are taken from cards in the possession of the Cambridge Anthropometrical Committee*, and are given in inches, but for Head Length and Head Breadth I have reduced the results to centimetres in order to facilitate comparison.

* Thanks are due to the Committee for their courtesy in allowing the use of their cards.

On Criminal Anthropometry

TABLE 8.

1000 *Cambridge Men.*

	S. D.	Mean
Head Length (cms.) ...	$\cdot 6161 \pm \cdot 0093$	$19\cdot 3509 \pm \cdot 0131$
Head Breadth (cms.)...	$\cdot 5055 \pm \cdot 0076$	$15\cdot 3959 \pm \cdot 0108$
Height (ins.)	$2\cdot 5447 \pm \cdot 0384$	$68\cdot 8550 \pm \cdot 0543$

TABLE 9.

1000 *Cambridge Men.*

	Coefficient of Correlation
Head Length and Head Breadth ...	$\cdot 3448 \pm \cdot 0188$
Height and Head Breadth	$\cdot 1529 \pm \cdot 0208$
Height and Head Length	$\cdot 2816 \pm \cdot 0196$

We will now calculate *D* and *S* for the 3000 criminals and 1000 Cambridge men :

TABLE 10.

1000 *Cambridge Men and 3000 Criminals.*

	S. D.		Mean	
	<i>D</i>	<i>S</i>	<i>D</i>	<i>S</i>
Head Length ...	$\cdot 0115$	$\cdot 0107$	$\cdot 1846$	$\cdot 0151$
Head Breadth ...	$\cdot 0041$	$\cdot 0088$	$\cdot 3518$	$\cdot 0124$
Height	$\cdot 0037$	$\cdot 0443$	$3\cdot 3195$	$\cdot 0626$

TABLE 11.

1000 *Cambridge Men and 3000 Criminals.*

	Coefficient of Correlation	
	<i>D</i>	<i>S</i>
Head Length and Head Breadth ...	$\cdot 0568$	$\cdot 0214$
Height and Head Breadth	$\cdot 0301$	$\cdot 0240$

We will next give the coefficient of variation, which is defined as the percentage variation in the mean, the standard deviation being treated as the total variation in the mean (*Phil. Trans.* Vol. 187, A., p. 277).

TABLE 12.

Coefficients of Variation. 1000 Cambridge Men and 3000 Criminals.

	1000 Cambridge Men	3000 Criminals
Head Length ...	3·1839 ± ·0481	3·1544 ± ·0275
Head Breadth ...	3·2836 ± ·0496	3·3332 ± ·0291
Height ...	3·6958 ± ·0558	3·8773 ± ·0338

The probable error of the coefficient of variation, v , is calculated from the formula:

$$\text{probable error} = \cdot 6745 \frac{v}{\sqrt{2n}} \left[1 + 2 \left(\frac{v}{100} \right)^2 \right]^{\frac{1}{2}},$$

where n is the number of measurements. (See Pearson, *Proceedings of the Royal Society*, Vol. 61, p. 345.)

From an examination of these results it appears that there is but little difference in variability or correlation between the criminal and the educated classes, but a most noteworthy difference in *means*. It is also to be noted that there is practically no difference in variability whether measured absolutely or by coefficients of variation.

Finally, we will compare the cephalic index in Cambridge men and criminals. In calculating the index it is to be noted (i) that 11 millimetres are deducted from the head measurements, in accordance with Dr Alice Lee's memoir "A First Study of the Correlation of the Human Skull" (*Phil. Trans.* Vol. 196, A., p. 252); and (ii) that we know only the mean Head Breadth and mean Head Length, and therefore in calculating the mean of the Cephalic Index, which is 100 times the ratio of Head Breadth and Head Length, we must apply the formulæ given in Professor Pearson's paper in the *Proceedings of the Royal Society*, Vol. 60, p. 492. He there shows that if x_1, x_2 , be the absolute sizes of any two correlated organs, m_1, m_2 , their means, σ_1, σ_2 , their standard deviations, r_{12} their coefficient of correlation, i_{12} the mean value of $\frac{x_1}{x_2}$, Σ_{12} the standard deviation of $\frac{x_1}{x_2}$, then

$$i_{12} = \frac{m_1}{m_2} (1 + v_2^2 - r_{12}v_1v_2),$$

$$\Sigma_{12} = i_{12} (v_1^2 + v_2^2 - 2r_{12}v_1v_2)^{\frac{1}{2}},$$

where

$$v_1 = \sigma_1/m_1, \quad v_2 = \sigma_2/m_2.$$

Applying these formulæ we find :

TABLE 13.

Cephalic Index. 1000 Cambridge Men and 3000 Criminals.

	S. D.		Mean	
	Head Index	Skull Index	Head Index	Skull Index
1000 Cambridge Men ...	2·9478	2·9021	79·562	78·330
3000 Criminals ...	2·7900	2·7435	78·538	77·227

The great difference in cephalic index between the two classes is to be noted, and it is interesting to observe that there is a large difference in the skull capacity as measured by Dr Lee's formula (*Phil. Trans.* Vol. 196, A., p. 235),

$$C = 6\cdot752 L + 11\cdot421 B - 1434\cdot06,$$

where C = capacity in cubic centims, L and B the length and breadth of skull in millims.

$$\begin{aligned} \text{For Criminals} \quad L &= 191\cdot663 - 11 = 180\cdot663 \\ B &= 150\cdot442 - 11 = 139\cdot442. \end{aligned}$$

$$\begin{aligned} \text{For Cambridge Men} \quad L &= 193\cdot509 - 11 = 182\cdot509 \\ B &= 153\cdot959 - 11 = 142\cdot959. \end{aligned}$$

Calculating C by the formula, we find it is

$$\begin{aligned} \text{For Criminals} \quad & 1378\cdot34 \\ \text{For Cambridge Men} \quad & 1430\cdot98 \end{aligned}$$

but in the absence of the measurement of the height of the skull, we do not lay much stress on these determinations.

Summing up the results of this part of the inquiry, I conclude that there is a substantial difference in stature, and in size and shape of head between the two classes; I do not assert that the source of the criminality is to be found in this difference, but only that criminals are drawn from a different section of the community. As bearing on this point it is worth noting that the mean height in Galton's middle-class measurements at the International Exhibition of 1884, viz. 67''·9, approaches our criminal mean more closely than does the Cambridge mean.

I should add that I believe the head measurements in the prisons and at Cambridge are made practically in the same way.

For further comparison I give the general position as to stature and cephalic index of our criminal classes.

TABLE 14.

Stature (Men).

	Number	Mean, cms.	Standard Deviation, cms.	Coefficient of Variation
Cambridge Students...	1000	174.88	6.46	3.70
English Sons* ...	1078	174.40	6.94	3.98
English Fathers* ...	1078	171.95	6.87	3.99
U.S.A. Recruits† ...	25878	170.94	6.56	3.84
N.S. Wales Criminals‡	2862	169.88	6.58	3.80
Frenchmen§ ...	284	166.80	6.47	3.88
English Criminals ...	3000	166.46	6.45	3.88
Germans§ ...	390	165.93	6.68	4.02

* Calculated from family measurement data, collected by Professor Pearson.

† Pearson: *Phil. Trans.*, Vol. 186, A., p. 386.

‡ Powys: *Biometrika*, Part 1. Vol. 1. p. 44.

§ Pearson: *The Chances of Death*, Vol. 1. p. 295.

TABLE 15.

Cephalic Index.

	Number	Mean	Standard Deviation
Bavarian Peasants* ...	100	83.41	3.58
Baden Recruits* ...	6748	81.15	3.63
French Peasants* ...	56	79.79	3.84
Cambridge Students ...	1000	78.33	2.90
Criminals ...	3000	77.23	2.74
Brahmans of Bengal* ...	100	76.86	3.65
Mahomedans of Bengal*	100	75.77	3.37
Whitechapel English* ...	107	74.73	3.31

* Professor Pearson's *The Chances of Death*, Vol. 1. p. 290. As his results are based either on skull measurements or on "corrected" head measurements, I have used the "corrected" figures of Table 13.

PART II.

APPLICATION OF METHODS.

(8) I will now proceed to apply the abbreviated method already described to the calculation of coefficients of correlation, standard deviations and means, and I will begin with Height and Head Breadth.

First, the following 16-fold table was formed (see Table II., p. 215).

3000 Criminals.

Height (feet and inches).

Head Breadth (cms.)		$5'4\frac{9}{16}$ and under	$5'4\frac{9}{16}$ — $5'5\frac{9}{16}$	$5'5\frac{9}{16}$ — $5'6\frac{9}{16}$	over $5'6\frac{9}{16}$	Totals
	$14'8$ and under	455	163	160	299	1077
	$14'9$ — 15 ...	158	71	71	151	451
	$15'1$ — $15'2$...	158	70	65	162	455
	over $15'2$...	283	158	162	414	1017
	Totals	1054	462	458	1026	3000

Forming from this a 4-fold table on the principle stated in § 3, we have:—

3000 Criminals.

Height (feet and inches).

Head Breadth (cms.)		$5'5\frac{9}{16}$ and under	over $5'5\frac{9}{16}$	Totals
	15 and under	847	681	1528
	over 15 ...	669	803	1472
	Totals	1516	1484	3000

Here $\alpha_1 = \cdot010667$ $\alpha_2 = \cdot018667$.

Hence from the probability integral tables:

$h = \cdot013366$ $k = \cdot023400$
 $\log HK = \bar{1} \cdot 2016623,$ $\epsilon = \cdot156822,$

and the equation in θ , where r , the coefficient of correlation, = $\sin \theta$, is:

$$\cdot156822 = \theta + \frac{1}{2}(\cdot000313)\theta^2 - (\cdot000727)\frac{\theta^3}{6} + (\cdot000313)\frac{5\theta^4}{24}.$$

$\theta = \cdot156822 = 8^\circ 59' 7''$ gives a very close solution, and therefore $r = \cdot1562$, the probable error being $\pm \cdot0190$.

On comparing this with the result obtained by the ordinary method, viz. $r = .1831 \pm .0119$, we conclude that we might improve our result by taking other limits of division; we therefore re-form our table as follows:

Height (feet and inches).

Head Breadth (cms.)		5'7 $\frac{3}{16}$ and under	over 5'7 $\frac{3}{16}$	Totals
	15'3 and under	1799	397	2196
	over 15'3 ...	587	217	804
	Totals	2386	614	3000

Here

$$\alpha_1 = .590667$$

$$\alpha_2 = .464000$$

$$h = .82507$$

$$k = .61889$$

$$\log HK = 2.9708265$$

$$\epsilon = .186974$$

and the equation in θ is:

$$.186974 = \theta + .255314\theta^2 - .133837\theta^3 + .044029\theta^4.$$

Approximating to the root by Newton's rule, we find:

$$\theta = .1794814 = 10^\circ 17' 1'' \text{ and } r = .1785,$$

a better result than before, but we will try yet another division, viz.

TABLE X.

Height (feet and inches).

Head Breadth (cms.)		5'4 $\frac{3}{16}$ and under	over 5'4 $\frac{3}{16}$	Totals
	14'8 and under	455	622	1077
	over 14'8 ...	599	1324	1923
	Totals	1054	1946	3000

Here

$$\alpha_1 = -.297333$$

$$\alpha_2 = -.282000$$

$$h = -.38173$$

$$k = -.36114$$

$$\log HK = 1.1420742$$

$$\epsilon = .184125.$$

The equation in θ now becomes:

$$.184125 = \theta + .068929\theta^2 - .042689\theta^3 + .024088\theta^4,$$

a solution of which is $\theta = .1820710 = 10^\circ 25' 55''$,

and

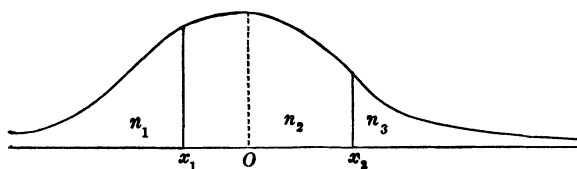
$$r = .1811$$

with a probable error of $\pm .0210$.

As this result coincides very closely with that obtained by the usual method, we will rest satisfied with it, and in subsequent tables adopt the divisions of

Height and Head Breadth on which it is based. Comparing it with the worst value obtained above, .1562, we see that even in that case the difference is only of the order .02, or of the probable error of sampling.

(9) I have next to find the standard deviations and means of the two characters; this I shall do by applying the method explained in Professor Pearson and Dr Lee's memoir "On the inheritance of characters not capable of exact quantitative measurement" (*Phil. Trans.* Vol. 195, A., p. 82). Let us suppose our 16-fold table reduced to a 9-fold table, and let n_1, n_2, n_3 , the three classes of one character which are thus formed, be represented by the areas of the normal curve in the accompanying diagram.



In all our investigations, the mean will fall within the area n_2 . Let $Ox_1 = p_1$, $Ox_3 = p_3$, be the distances of the mean from the boundaries of n_2 ; then $p_1 + p_3$ in absolute measurement is known, = η , say, being the range of the class n_2 .

Remembering that the equation to the normal curve is of the form

$$y = \frac{N}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\frac{x^2}{\sigma^2}},$$

where N is the total frequency and σ the S.D., we have

$$\frac{n_2 + n_3 - n_1}{N} = \sqrt{\frac{2}{\pi}} \int_0^{h_1} e^{-\frac{1}{2}x^2} dx,$$

$$\frac{n_1 + n_2 - n_3}{N} = \sqrt{\frac{2}{\pi}} \int_0^{h_3} e^{-\frac{1}{2}x^2} dx,$$

where

$$h_1 = \frac{p_1}{\sigma}, \quad h_3 = \frac{p_3}{\sigma}.$$

Now the left-hand side is known, and from a table of the areas of the normal curve h_1 and h_3 are found. Therefore as $\sigma = \frac{p_1 + p_3}{h_1 + h_3} = \frac{\eta}{h_1 + h_3}$, the standard deviation is known. Also, as $p_1 = h_1\sigma$, the distance of the mean from the left-hand boundary of n_2 , and therefore the mean itself, are known.

To find the probable error of σ , we have

$$\sigma = \frac{\eta}{h_1 + h_3}, \quad \therefore \delta\sigma = -\frac{\eta(\delta h_1 + \delta h_3)}{(h_1 + h_3)^2}.$$

$$\therefore \Sigma^2_\sigma = \frac{\eta^2}{(h_1 + h_3)^4} (\Sigma^2_{h_1} + \Sigma^2_{h_3} + 2\Sigma_{h_1}\Sigma_{h_3}R_{h_1,h_3}).$$

Now

$$\frac{N - 2n_1}{2N} = \frac{1}{\sqrt{2\pi}} \int_0^{h_1} e^{-\frac{1}{2}x^2} dx;$$

therefore, if

$$H = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}h^2},$$

we have

$$-\delta n_1 = NH_1 \delta h_1 \text{ and } \Sigma_{h_1} = \Sigma_{n_1} / (NH_1).$$

Similarly

$$-\delta n_3 = NH_3 \delta h_3 \text{ and } \Sigma_{h_3} = \Sigma_{n_3} / (NH_3).$$

Also $\Sigma_{h_1} \Sigma_{h_3} R_{h_1 h_3} = \Sigma_{n_1} \Sigma_{n_3} R_{n_1 n_3} / N^2 H_1 H_3$, and it is shown in Professor Pearson's memoir "On the correlation of characters not quantitatively measurable" (*Phil. Trans.* Vol. 195, A., p. 11), that

$$\Sigma_{n_1}^2 = \frac{n_1(N - n_1)}{N}, \quad \Sigma_{n_3}^2 = \frac{n_3(N - n_3)}{N},$$

$$\Sigma_{n_1} \Sigma_{n_3} R_{n_1 n_3} = -\frac{n_1 n_3}{N},$$

therefore

$$\Sigma_{h_1}^2 = \frac{n_1(N - n_1)}{N^2 H_1^2}, \quad \Sigma_{h_3}^2 = \frac{n_3(N - n_3)}{N^2 H_3^2},$$

$$2\Sigma_{h_1} \Sigma_{h_3} R_{h_1 h_3} = -\frac{2n_1 n_3}{N^2 H_1 H_3}.$$

The probable error of σ is accordingly $\cdot 67449 \Sigma_{\sigma}$,

$$= \cdot 67449 \frac{\eta}{(h_1 + h_3)^2} \left\{ \frac{n_1(N - n_1)}{N^2 H_1^2} + \frac{n_3(N - n_3)}{N^2 H_3^2} - \frac{2n_1 n_3}{N^2 H_1 H_3} \right\}^{\frac{1}{2}}.$$

We have next to find the probable error in p_1 , that is, the probable error of the mean.

$$p_1 = h_1 \sigma,$$

$$\delta p_1 = h_1 \delta \sigma + \sigma \delta h_1.$$

Therefore

$$\Sigma_{p_1}^2 = h_1^2 \Sigma_{\sigma}^2 + \sigma^2 \Sigma_{h_1}^2 + 2h_1 \sigma \Sigma_{\sigma} \Sigma_{h_1} R_{\sigma h_1};$$

also

$$h_1 + h_3 = \frac{\eta}{\sigma},$$

$$\delta h_3 = -\delta h_1 - \eta \frac{\delta \sigma}{\sigma^2},$$

$$\Sigma_{h_3}^2 = \Sigma_{h_1}^2 + \frac{\eta^2}{\sigma^4} \Sigma_{\sigma}^2 + \frac{2\eta}{\sigma^2} \Sigma_{\sigma} \Sigma_{h_1} R_{\sigma h_1}.$$

Substituting for $\Sigma_{\sigma} \Sigma_{h_1} R_{\sigma h_1}$ from the first equation, and remembering that $\eta = \sigma(h_1 + h_3)$, we have finally, after a few reductions,

$$\Sigma_{p_1}^2 = \frac{\sigma^2(h_1 \Sigma_{h_3}^2 + h_3 \Sigma_{h_1}^2)}{h_1 + h_3} - \Sigma_{\sigma}^2 h_1 h_3,$$

and the probable error = $\cdot 67449 \Sigma_{p_1}$.

(10) We will now apply these formulæ.

Height and Head Breadth. We first form the 9-fold Table XI. (p. 222).

Taking Height first, we have :

$$\begin{aligned} n_1 &= 1054, & n_2 &= 920, & n_3 &= 1026. \\ h_1 &= \cdot 38173, & h_3 &= \cdot 40702, & \eta &= 2. \end{aligned}$$

Therefore
$$\sigma = \frac{2}{\cdot 78875} = 2\cdot 53566 \text{ inches,}$$

$$p_1 = \sigma \times h_1 = \cdot 967937.$$

The mean is therefore

$$64\frac{9}{16} + \cdot 967937 = 65\cdot 5304 \text{ inches.}$$

For Head Breadth

$$\begin{aligned} n_1 &= 1077, & n_2 &= 906, & n_3 &= 1017, \\ h_1 &= \cdot 36114, & h_3 &= \cdot 415198, & \eta &= \cdot 4, \end{aligned}$$

$$\sigma = \frac{\cdot 4}{\cdot 77634} = \cdot 515238 \text{ (cm.),}$$

$$p_1 = \sigma \times h_1 = \cdot 186073.$$

In finding the mean it is to be noted that the lower boundary of n_2 is 14·85 as the measurements are taken to the nearest millimetre ; the mean is therefore

$$14\cdot 85 + \cdot 186073 = 15\cdot 0361 \text{ cm.}$$

The probable errors of the S.D. are

$$\text{Height} \quad \pm \cdot 0495$$

$$\text{Head Breadth} \quad \pm \cdot 01015.$$

Height and Left Middle Finger. Our 4-fold table is Table XII., p. 222 (see also Table III., p. 216).

$$h = -\cdot 38173, \quad k = \cdot 01169, \quad \epsilon = \cdot 714334.$$

The equation in θ is :

$$\cdot 714334 = \theta - \cdot 002232\theta^2 - \cdot 024306\theta^3 - \cdot 000849\theta^4 - \cdot 000919\theta^5,$$

whence

$$\theta = \cdot 7251964, \quad r = \cdot 6633.$$

The 9-fold table is Table XIII., p. 222 (see also Table III., p. 216).

For Finger

$$\begin{aligned} n_1 &= 706, & n_2 &= 1612, & n_3 &= 682, \\ h_1 &= \cdot 7214, & h_3 &= \cdot 74769, & \eta &= \cdot 8, \end{aligned}$$

$$\sigma = \frac{\cdot 8}{1\cdot 46909} = \cdot 544555,$$

$$p_1 = h_1\sigma = \cdot 392842.$$

Therefore the mean is

$$11.15 + .392842 = 11.542842,$$

and the probable error of $\sigma = .0075$.

Head Length and Head Breadth. The 4-fold table is Table XIV., p. 222 (see also Table I., p. 214).

$$h = -.03761; k = -.36114; \epsilon = .408616.$$

$$.408619 = \theta + .006791\theta^2 - .021942\theta^3 + .002606\theta^4 - .008325\theta^5,$$

whence

$$\theta = .4090070, \quad r = .3977.$$

For s.d. of Head Length, we have from the 9-fold Table XV., p. 222,

$$n_1 = 463, \quad n_2 = 2043, \quad n_3 = 494,$$

$$h_1 = 1.018019, \quad h_3 = .975451, \quad \eta = 1.2.$$

$$\sigma = \frac{1.2}{1.99347} = .6019654,$$

and

$$p_1 = h_1\sigma = .6128122,$$

therefore

$$\text{Mean} = 18.55 + .6128122 = 19.1628122,$$

and the probable error of σ is .0071.

(11) *Comparison of Results.* We can now compare the results obtained by the new method with those obtained from the full correlation tables:

TABLE 16.

Coefficient of Correlation.

	Old Method	New Method
Head Length and Head Breadth	.4016 $\pm$.0103	.3977 $\pm$.0176
Head Breadth and Height ...	.1831 $\pm$.0119	.1811 $\pm$.0210
L. M. Finger and Height ...	.6608 $\pm$.0069	.6633 $\pm$.0142

TABLE 17.

S.D. and Mean.

	Standard Deviation		Mean	
	Old Method	New Method	Old Method	New Method
Head Length (cm.)	.6046 $\pm$.0053	.6020 $\pm$.0071	19.1663 $\pm$.0075	19.1628 $\pm$.0086
Head Breadth (cm.)	.5014 $\pm$.0044	.5152 $\pm$.0101	15.0442 $\pm$.0062	15.0361 $\pm$.0072
Finger (cm.) ...	.5479 $\pm$.0048	.5446 $\pm$.0075	11.5474 $\pm$.0068	11.5428 $\pm$.0075
Height (ins.) ...	2.5410 $\pm$.0221	2.5357 $\pm$.0495	65.5355 $\pm$.0313	65.5304 $\pm$.0352

The extremely close agreement of the results thus obtained in two very different ways, especially the agreement in the means, is deserving of special attention and is a striking illustration of the value of the new method.

(12) *Application of Methods, continued.* I will now proceed to calculate the remaining standard deviations, means, and coefficients of correlation.

Head Length and Face Breadth. See Tables XVI. and XVII., p. 223.

$$h = -\cdot03761; \quad k = 0; \quad \epsilon = \cdot405552,$$

$$\cdot405552 = \theta - \cdot001415 \frac{\theta^2}{6};$$

whence $\theta = \cdot4055676$ and $r = \cdot3945 \pm \cdot0172$.

For S.D. and Mean of Face Breadth:

$$n_1 = 1039, \quad n_2 = 922, \quad n_3 = 1039, \quad h_1 = h_3 = \cdot39524;$$

therefore

$$\sigma = \cdot5060216 \pm \cdot009856.$$

$$p_1 = h_1\sigma = \cdot2000, \text{ therefore Mean} = 13\cdot45 + \cdot2 = 13\cdot65.$$

Head Length and Cubit. Tables XVIII. and XIX., p. 223.

$$h = -\cdot03761, \quad k = -\cdot067734, \quad \epsilon = \cdot310452,$$

$$\cdot310452 = \theta + \cdot001274\theta^2 - \cdot001\theta^3 + \cdot000529\theta^4;$$

whence

$$\theta = \cdot3103547; \quad r = \cdot3054.$$

For S.D. and Mean of Cubit:

$$n_1 = 911, \quad n_2 = 1002, \quad n_3 = 1087, \quad h_1 = \cdot51389, \quad h_3 = \cdot35224,$$

whence

$$\sigma = 1\cdot96275 \pm \cdot0171.$$

Also $p_1 = h_1\sigma = 1\cdot00864$; therefore Mean = $44\cdot05 + p_1 = 45\cdot05864$.

Head Length and Foot. Tables XX. and XXI., pp. 223, 224.

$$h = -\cdot03761, \quad k = -\cdot10295, \quad \epsilon = \cdot345875,$$

$$\cdot345875 = \theta + \cdot001936\theta^2 - \cdot002\theta^3 + \cdot000801\theta^4;$$

whence

$$\theta = \cdot3457144, \quad r = \cdot3389.$$

For S.D. and Mean of Foot:

$$n_1 = 883, \quad n_2 = 979, \quad n_3 = 1138, \quad h_1 = \cdot540778, \quad h_3 = \cdot307234,$$

whence

$$\sigma = 1\cdot1792286 \pm \cdot0103.$$

Also $p_1 = h_1\sigma = \cdot6377009$; therefore Mean = $25\cdot05 + p_1 = 25\cdot6877$.

The divisions chosen for Foot are not quite satisfactory, the class "over 26" being too large.

Having now found the S.D. and Mean of all seven characters, we no longer require 9-fold tables, and we shall therefore now use only 4-fold tables in order to find r .

Head Length and Finger. Table XXII., p. 224.

$$h_1 = -.03761, \quad k = .01169, \quad \epsilon = .305411, \\ .305411 = \theta - .00022\theta^2 - .000259\theta^3 - .000091\theta^4;$$

whence $\theta = .3054401, \quad r = .3007 \pm .0181.$

Head Length and Height. Table XXIII., p. 224.

$$h = -.38173, \quad k = -.03761, \quad \epsilon = .34673, \\ .34673 = \theta + .007178\theta^2 - .024479\theta^3 + .002727\theta^4;$$

whence $\theta = .3468384, \quad r = .3399 \pm .0207.$

Head Breadth and Face Breadth. Table XXIV., p. 224.

$$h = -.36114, \quad k = 0, \quad \epsilon = .660597, \\ .660597 = \theta - .021737\theta^3 + .008269\theta^5;$$

whence $\theta = .6659329, \quad r = .6178.$

If we calculate r from a division nearly median (Table XXIV.^a, p. 224) we have:

$$h = .0234, \quad k = 0, \quad \epsilon = .655725, \\ .655725 = \theta - .0000913\theta^3 - .0000365\theta^5;$$

whence $\theta = .655755, \quad r = .6098.$

Head Breadth and Finger. Table XXV., p. 225.

$$h = -.36114, \quad k = .01169, \quad \epsilon = .15085, \\ .15085 = \theta - .002111\theta^2 - .021757\theta^3 - .000811\theta^4;$$

whence $\theta = .1509732, \quad r = .1504.$

If we adopt a median division (Table XXV.^a, p. 225) we shall find $r = .1459.$

Head Breadth and Cubit. Table XXVI., p. 225.

$$h = -.36114, \quad k = -.06773, \quad \epsilon = .135737, \\ .135737 = \theta + .012230\theta^2 - .022402\theta^3 + .004684\theta^4;$$

whence $\theta = .1355670, \quad r = .1352 \pm .0196.$

Head Breadth and Foot. Table XXVII., p. 225.

$$h = -.10295, \quad k = -.36114, \quad \epsilon = .208238, \\ .208238 = \theta + .01859\theta^2 - .023273\theta^3 + .007092\theta^4;$$

whence $\theta = .2076318, \quad r = .2061.$

Face Breadth and Finger. Table XXVIII., p. 225.

$$h = 0, \quad k = .01169, \quad \epsilon = .326748, \\ .326748 = \theta - .000023\theta^3,$$

whence $\theta = .3267488, \quad r = .3210.$

Face Breadth and Cubit. Table XXIX., p. 226.

$$h = 0, \quad k = -\cdot 06773, \quad \epsilon = \cdot 292839, \\ \cdot 292839 = \theta - \cdot 0007647\theta^2,$$

whence $\theta = \cdot 2928580, \quad r = \cdot 2887.$

Face Breadth and Foot. Table XXX., p. 226.

$$h = 0, \quad k = -\cdot 102952, \quad \epsilon = \cdot 371625, \\ \cdot 371625 = \theta - \cdot 0017665\theta^2,$$

whence $\theta = \cdot 3717160, \quad r = \cdot 3632.$

Face Breadth and Height. Table XXXI., p. 226.

$$h = 0, \quad k = -\cdot 38173, \quad \epsilon = \cdot 351419, \\ \cdot 351419 = \theta - \cdot 024286\theta^2 - \cdot 009185\theta^3;$$

whence $\theta = \cdot 3525333, \quad r = \cdot 3453.$

Foot and Finger. Table XXXII., p. 226.

$$h = -\cdot 10295, \quad k = \cdot 01169, \quad \epsilon = \cdot 859268, \\ \cdot 859268 = \theta - \cdot 000602\theta^2 - \cdot 001789\theta^3 - \cdot 000249\theta^4 - \cdot 000713\theta^5,$$

whence $\theta = \cdot 8613318, \quad r = \cdot 7587.$

Finger and Cubit. Table XXXIII., p. 226.

$$h = \cdot 01169, \quad k = -\cdot 06773, \quad \epsilon = 1\cdot 007431, \\ 1\cdot 007431 = \theta - \cdot 000396\theta^2 - \cdot 000787\theta^3 - \cdot 000164\theta^4 - \cdot 000314\theta^5;$$

whence $\theta = 1\cdot 0091435, \quad r = \cdot 8464 \pm \cdot 0079.$

As the cubit includes the finger, their correlation may be considered "spurious." Let us define "forearm" as the cubit minus the finger; then we can find S.D. and mean of forearm, and correlation of forearm and finger, as follows:—

Let a = forearm, c = cubit, f = finger.

$$c = a + f \text{ and } cf = af + f^2.$$

Therefore $S(cf) = S(af) + S(f^2),$

or $r_{fc} \sigma_f \sigma_c = r_{af} \sigma_a \sigma_f + \sigma_f^2,$

therefore $r_{af} = \frac{r_{fc} \sigma_c - \sigma_f}{\sigma_a},$

which gives the correlation of forearm and finger. Substituting the known values of the symbols on the right-hand side we have:

$$r_{af} = \cdot 7297.$$

Also $a = c - f$, therefore $\sigma_a^2 = \sigma_c^2 + \sigma_f^2 - 2\sigma_c\sigma_f r_{cf}$,
 and substituting we find $\sigma_a = 1.5297$,
 and $\text{Mean}_a = \text{Mean}_c - \text{Mean}_f = 33.5158$.

Left Foot and Left Cubit. Table XXXIV., p. 227.

$h = -.10295$, $k = -.06773$, $\epsilon = .923660$,
 $.923660 = \theta + .003486\theta^2 - .002523\theta^3 + .001439\theta^4 - .0009986\theta^5$;
 whence $\theta = .9222992$; $r = .79699$.

Height and Left Cubit. Table XXXV., p. 227.

$h = .01337$, $k = -.06773$, $\epsilon = .925690$,
 $.925690 = \theta - .000453\theta^2 - .000794\theta^3 - .000188\theta^4 - .000317\theta^5$;
 whence $\theta = .9270676$; $r = .7999$.

Here we have adopted the median division. If we work with the skew division, Table XXXV.^a, p. 227, we have:

$h = -.38173$, $k = -.06773$, $\epsilon = .910952$,
 $.910952 = \theta + .012927\theta^2 - .024939\theta^3 + .004901\theta^4 - .009349\theta^5$;
 whence $\theta = .9222068$; $r = .7969$;

but it will be observed that in the second equation for θ the terms converge much more slowly than in the first; we will therefore retain the first value, especially as the difference between the two is only .00292.

Height and Left Foot. Table XXXVI., p. 227.

$h = -.013366$, $k = -.102952$, $\epsilon = .82577$,
 $.825770 = \theta - .000688\theta^2 - .001796\theta^3 - .000285\theta^4 - .0007153\theta^5$;
 whence $\theta = .8276709$; $r = .7364$.

Here again we have adopted the median division.

Adopting the skew division, Table XXXVI.^a, p. 227, we have:

$h = -.38173$, $k = -.10295$, $\epsilon = .881436$,
 $.881436 = \theta + .01965\theta^2 - .025795\theta^3 + .007422\theta^4 - .009565\theta^5$;
 whence $\theta = .8845487$; $r = .7736$.

It will be noticed that when r is large, the terms involving the higher powers of θ do not always converge as rapidly as we should like in the equations for θ ; the values of θ are therefore not to be relied on beyond the 4th figure or so. To go up to the 6th power of θ would considerably increase the labour of solving the equations; we have therefore not gone beyond the 5th power, the term involving which is:

$$+ [h^4k^4 + (h^2 + k^2)\{3(h^2 + k^2) - 6h^2k^2 - 8\} + 20h^2k^2] \frac{\theta^5}{120}.$$

(13) For convenience of reference, we will now collect all the standard deviations, means, and coefficients of correlation as ascertained in the course of the preceding inquiry; also the coefficients of variation.

TABLE 18.
3000 *Criminals*.

	Standard Deviation	Coefficient of Variation	Mean	
Head Length (cms.) ...	·6046	3·1544	19·1663	Table 1
Head Breadth (cms.) ...	·5014	3·3332	15·0442	Table 1
Face Breadth (cms.) ...	·5060	3·7071	13·6500	§ 12
Left Middle Finger (cms.)	·5479	4·7445	11·5474	Table 1
Left Cubit (cms.) ...	1·9627	4·3560	45·0586	§ 12
Left Foot (cms.) ...	1·1792	4·5906	25·6877	§ 12
Height (ins.) ...	2·5410	3·8773	65·5355	Table 1
„ (cms.) ...	6·4540		166·4572	

TABLE 19.
Coefficients of Correlation. 3000 Criminals.

	Head Length	Head Breadth	Face Breadth	Finger	Cubit	Foot	Height
Head Length	1·	·40163	·39454	·30071	·30539	·33886	·33993
Head Breadth	·40163	1·	·61779	·15040	·13515	·20614	·18308
Face Breadth	·39454	·61779	1·	·32097	·28869	·36322	·34527
Finger ...	·30071	·15040	·32097	1·	·84638	·75871	·66084
Cubit ...	·30539	·13515	·28869	·84638	1·	·79699	·79986
Foot ...	·33886	·20614	·36322	·75871	·79699	1·	·73636
Height ...	·33993	·18308	·34527	·66084	·79986	·73636	1·

(14) It now becomes necessary to test the probable errors of some of these results.

TABLE 20.
Probable Errors of the Means in Criminals.

	By Full Correlation Table of 3000	By Full Correlation Table of 1306	By New Method	$h_1 + h_2$
Head Length ...	·0075	·0112	·0086	1·99347
Head Breadth ...	·0062	·0096	·0072	·77634
Face Breadth ...	—	·0094	·0070	·79048
Finger ...	·0067	·0102	·0075	1·46909
Height ...	·0313	·0484	·0352	·78875

TABLE 21.

Probable Errors of S.D. in Criminals.

	By Full Correlation Table, 3000	By Full Correlation Table, 1306	By New Method, 3000	$h_1 + h_3$
Head Length (cms.)	·0053	·0079	·0071	1·99347
Head Breadth (cms.)	·0044	·0068	·0101	·77634
Face Breadth (cms.)	—	·0066	·0099	·79048
Finger (cms.) ...	·0048	·0072	·0075	1·46909
Height (ins.) ...	·0221	·0342	·0495	·78875

TABLE 22.

Probable Errors of r in Criminals.

	By New Method	By Formula $\frac{.6745 (1 - r^2)}{\sqrt{1100}}$
Height and Head Breadth ($r = .1831$)	·0210	·0197
Head Length and Face Breadth ($r = .3945$)	·0172	·0172
Head Length and Finger ($r = .3007$)	·0181	·0185
Head Length and Height ($r = .3399$)	·0207	·0180
Head Breadth and Cubit ($r = .1352$)	·0196	·0200
Finger and Cubit ($r = .8464$)	·0079	·0058

It will be noted from the results for Head Length and Finger that the Probable Errors of the standard deviations by the new method for 3000 are very nearly equal to those by the usual method for 1306 when $h_1 + h_3$ is large (see § 9), that is, when we arrange our 9-fold table so that the middle division is large. A reference to § 11 will also show that by arranging our tables in this way we obtained remarkably good results for s.d. and mean of Head Length and Finger; we might therefore expect improved results if we were to arrange all our 9-fold tables throughout on this principle.

It is to be noted that the Probable Errors of the means by the new method are practically as small as those of the standard deviations.

Owing to the laborious character of the calculation for finding the probable error of r by the new method, I have worked out only eight of the 21 errors (see Tables 16 and 22), but these eight being fairly representative will give a good idea of the magnitudes involved.

In the 2nd column of Table 22, the probable error is calculated by the usual formula for normal frequency, supposing $n = 1100$, and on the whole the results correspond very closely with those obtained by the new method, except in the case of the last coefficient, where the difference is considerable, but as it happens the absolute amount of error is small in either case. We may therefore

conclude that by applying the new method to 3000 cases we shall obtain the coefficients of correlation to the same degree of accuracy as if we calculated them from a full correlation table of 1100 cases.

As a result of the preceding investigation I conclude that the new method (i) saves a great deal of time and labour; (ii) gives excellent results for standard deviation and mean, especially if the middle division of the 9-fold table is made comparatively large; and (iii) gives less good results for r when r is high, owing to the slowness of convergence of the terms in the equation for θ , but on the whole it enables us to calculate r with an accuracy fairly comparable to that attained from an ordinary frequency table containing about one-third the number of frequencies.

(15) I will now state some results, based on Tables 18 and 19, which I think are sufficiently novel and important to be emphasised. I note that the determinations of Left Foot, so far as I can ascertain, are the first that have been made, and have therefore the interest of novelty.

The following table shows how the mean head measurements of criminals compare with those of the British Association and the University College series which are given by Dr Alice Lee on page 251 of her memoir, "A first study of the Correlation of the Human Skull" (*Phil. Trans.*, Vol. 196); the B.A. measurements are averages obtained by Dr Lee from the values given for several years in the "Reports of the B.A. Anthropometric Committee"; the University College measurements were taken by Professor Karl Pearson on twenty-five members of the College staff.

TABLE 23.

Mean Head Measurements.

	B. A.	U. C. Staff	Criminals
Length	198.1	196.38	191.7
Breadth	155.0	153.48	150.4

As regards variability, Dr Lee's memoir, p. 230, enables me to make the following comparison:

TABLE 24.

Coefficients of Variation.

	Head Length	Head Breadth
Aino Males ...	3.195	2.759
German Males ...	3.371	3.887
Criminals ...	3.154	3.333

And if we compare our coefficients of variation as a whole with those obtained for other parts of the body by various investigators, we shall find a general agreement in the results.

For instance, Miss Whiteley has shown that the coefficients of variation for the first joints of the fingers range from 4.695 to 5.361 (*Proceedings of the Royal Society*, Vol. 65, p. 129); and Professor Pearson has found that these coefficients for the long bones (Aino and French) range from 4.655 to 5.425.

From these figures it appears that 3 to 5.5 are representative values for variability in man, while in plants it may run to 40!

I may direct attention to some of the general results of the table of correlation (Table 19); e.g., the close relationship of finger and foot, of cubit and foot, which I think is novel and interesting, of height and finger, and cubit and finger. It will be noticed that the high correlations between finger, cubit, foot and height agree generally with the correlations between height and long bones as calculated by Professor Pearson (*Phil. Trans.* Vol. 192, A., p. 181, Table III.). For instance, he gives the coefficient of correlation between height and radius as .6956, as compared with .7999 the coefficient of correlation between height and cubit in criminals.

Turning now to the head measurements, Head Length, Head Breadth and Face Breadth, we observe that our results confirm the view that parts of the head are less highly correlated together than other parts of the body. It is also to be noticed that comparatively little correlation exists between the head and the other four characters, and that Head Length is more closely correlated with Cubit than is Head Breadth or Face Breadth, while Face Breadth is more nearly correlated with Foot than is Head Length or Head Breadth. Long head associated with long arm (i.e. cubit) rather suggests the gorilla type, but I can think of no type suggested by the association of long foot and broad face.

PART III.

APPLICATION OF THESE RESULTS TO CRIMINAL IDENTIFICATION.

(16) The conditions to be satisfied in selecting a series of organs for criminal identification are:

- (i) Comparative ease and accuracy of measurement.
- (ii) Small correlation between them, so that fairly few organs will provide a reliable index to the criminal population.

Head Length, Head Breadth and Face Breadth can be fairly accurately determined; Cubit and Finger with less accuracy; Height changes about $\frac{1}{8}$ " per 10 years from about 27 years of age*, and is not very reliable; Foot is a doubtful measurement.

* *Biometrika*, Vol. I. p. 49.

Now the organs selected by Scotland Yard are very far from being slightly correlated. We have seen that Face Breadth and Head Breadth are highly correlated, and the correlations between Finger, Cubit, Foot and Height are very high. It cannot therefore be said that these seven measurements form an ideal group, and I do not suppose they would have been chosen if their correlations had been actually known beforehand. My results at this point conflict with those of Dr Garson, who writes in his paper in the *Journal of the Anthropological Institute* to which I have already referred: "In a mixed population such as we have to deal with in England, the correlation between the different measurements used for the classification of criminals is slight."

Supposing, however, these organs have been selected, we may ask in the next place: What is the best order for entering an Index? This is the problem to which we shall next turn, and for its solution I propose to apply a general theorem in correlation which may be stated thus:

(17) Given n variables, then the most probable value A_1 of one of the variables, for given values $A_2, A_3, \dots A_n$, of the remaining $n - 1$, is given by the equation

$$A_1 - m_1 = -\sigma_1 \left\{ \frac{R_{12}}{R_{11}} \frac{(A_2 - m_2)}{\sigma_2} + \frac{R_{13}}{R_{11}} \frac{(A_3 - m_3)}{\sigma_3} + \dots + \frac{R_{1n}}{R_{11}} \frac{(A_n - m_n)}{\sigma_n} \right\},$$

and its variability $\bar{s}_1 = \sigma_1 \sqrt{\frac{\Delta}{R_{11}}}$, where σ_1 is the S.D. of the organ A_1 , and σ_q of the organ A_q ; m_1 the mean of A_1 and m_q the mean of A_q ; r_{1q} the correlation coefficient of A_1 and A_q , and $r_{qq'}$ the correlation coefficient of A_q and $A_{q'}$; and finally Δ is the determinant

$$\begin{vmatrix} 1 & r_{12} & r_{13} & r_{14} & \dots & r_{1n} \\ r_{21} & 1 & r_{23} & . & & . \\ \vdots & . & . & . & & . \\ r_{n1} & r_{n2} & . & . & & 1 \end{vmatrix}$$

and R_{pq} the minor corresponding to r_{pq} .

When $A_2, A_3, \dots A_n$ are known for only one individual, the probable error of the determination A_1 is

$$.67449 \sigma_1 \sqrt{\frac{\Delta}{R_{11}}};$$

when they are known for p individuals, the probable error of A_1 , which is then the corresponding mean value, becomes

$$.67449 \sigma_1 \sqrt{\frac{\Delta}{p R_{11}}}.$$

When $n = 2$, the reconstruction formula becomes

$$A_1 - m_1 = \frac{\sigma_1}{\sigma_2} r_{12} (A_2 - m_2),$$

and the probable error

$$.67449 \sigma_1 \sqrt{\frac{1-r_{12}^2}{p}}.$$

(See Pearson, *Phil. Trans.* Vol. 192, A., pp. 171-2.)

(18) In our case Δ is the determinant of the 7th order* :

1·	·40163	·39454	·30071	·30539	·33886	·33993
·40163	1·	·61779	·15040	·13515	·20614	·18308
·39454	·61779	1·	·32097	·28869	·36322	·34527
·30071	·15040	·32097	1·	·84638	·75871	·66084
·30539	·13515	·28869	·84638	1·	·79699	·79986
·33886	·20614	·36322	·75871	·79699	1·	·73636
·33993	·18308	·34527	·66084	·79986	·73636	1·

the value of which is found by the laborious process of reduction to be = ·012129.

Also	$R_{11} = .016272$	and	$\sqrt{\Delta/R_{11}} = .86336$
	$R_{22} = .020907$		$\sqrt{\Delta/R_{22}} = .76167$
	$R_{33} = .022556$		$\sqrt{\Delta/R_{33}} = .73330$
	$R_{44} = .047496$		$\sqrt{\Delta/R_{44}} = .50534$
	$R_{55} = .071864$		$\sqrt{\Delta/R_{55}} = .41083$
	$R_{66} = .040198$		$\sqrt{\Delta/R_{66}} = .54930$
	$R_{77} = .038347$		$\sqrt{\Delta/R_{77}} = .56240.$

Now the best organ to leave to the last will be that the variability of which is least reduced by selecting the other six. For this means that the individuals from whom we have in the last instance to select our man, will be least crowded together and therefore least likely to be indistinguishable. As $\bar{s} = \sigma \sqrt{\frac{\Delta}{R}}$, the greater $\sqrt{\frac{\Delta}{R}}$, the less the reduction in variability; this function we have already calculated for $n=7$, and we see that in that case it is greatest for Head Length. We will therefore keep Head Length to the last, and proceed to calculate $\sqrt{\frac{\Delta}{R}}$ for the six organs that remain after the elimination of Head Length. The new Δ is obviously $R_{11} = .016272$, and the new R 's can be calculated without much trouble. On comparing the new $\sqrt{\frac{\Delta}{R}}$ series, we shall find that Head Breadth is the next character to be eliminated, and we proceed in like manner until we have dealt with all the characters.

* Values of the correlation coefficients are given to 5 places of figures in order to calculate the determinants more accurately.

The result of these operations is shown in the following table :

TABLE 25.

Showing the value of $\sqrt{\Delta/R}$ for 3, 4, 5, 6 and 7 characters.

	7	6	5	4	3
Head Length ...	·86336	—	—	—	—
Head Breadth ...	·76167	·78408	—	—	—
Face Breadth ...	·73330	·73763	·91541	—	—
Finger ...	·50534	·50568	·50607	·50959	·51403
Cubit ...	·41083	·41083	·41089	·41452	·47660
Foot ...	·54930	·55016	·55060	·55671	·58296
Height ...	·56240	·56502	·56504	·57244	—

The order in which the various characters should be taken is therefore: 1st, Cubit; 2nd, Finger; 3rd, Foot; 4th, Height; 5th, Face Breadth; 6th, Head Breadth; 7th, Head Length, and is the same as that which is given by the first column, for $n = 7$. It is to be noted that the head measurements come last.

This result conflicts with the practice at Scotland Yard, where, I understand, the cabinet is entered in the order Head Length, Head Breadth, Face Breadth, Finger, Cubit, Foot and Height, the head measurements being taken first.

Should it be maintained that in fixing the order of the characters we should use $\bar{s}$, and not $\sqrt{\frac{\Delta}{R}}$, as the criterion, that is, that we should leave to the last the organ whose absolute variability is greatest, not the organ whose variability is least reduced, after selection of the other six, I have calculated the following table to show how this principle will work out :

TABLE 26.

Showing the value of $\bar{s} = \sigma \sqrt{\Delta/R}$ for 3, 4, 5, 6 and 7 characters.

	7	6	5	4	3
Head Length $\sigma = \cdot6046$	·5220	·5244	·5251	·5289	—
Head Breadth $\sigma = \cdot5015$	·3819	·3821	·3823	·3823	·3935
Face Breadth $\sigma = \cdot5060$	·3711	·3738	·3741	·3769	·3804
Finger $\sigma = \cdot5479$	·2769	·2801	·3543	·5051	·5178
Cubit $\sigma = 1\cdot9628$	·8064	·9323	—	—	—
Foot $\sigma = 1\cdot1792$	·6478	·6707	·7478	—	—
Height $\sigma = 6\cdot4540$	3·6297	—	—	—	—

In the case of height the unit of record is $\frac{1}{8}$ th inch = 3·175 mm., for the other characters the unit is 1 mm.; we therefore divide the 3·62971 of the table by

3.175 to obtain a true comparison for height, and find $\bar{s} = 1.1432$, which is still the greatest value of the 7 column.

On this principle, then, the order is 1st, Face Breadth; 2nd, Head Breadth; 3rd, Finger; 4th, Head Length; 5th, Foot; 6th, Cubit; and 7th, Height; it will be observed that this is the order of the standard deviations of the characters, but differs widely from the Scotland Yard order.

I think, however, that $\sqrt{\Delta/R}$ is the correct criterion. We cannot read height to the same accuracy as head length. $\bar{s}$ for height = 3.6297, and for head length .5220, but head length is read to $\frac{1}{2}$ mm. and height to $\frac{1}{16}$ in. Now head length is hardly likely to be correct to more than 1 mm. or height to $\frac{1}{4}$ inch = 6 mm. say, or even to $\frac{1}{2}$ inch = 12 mm.; hence I doubt if stature with an $\bar{s}$ of 3.6 is really better than head length with a range of .5. The right principle seems to be to suppose each organ on its own S.D. equally useful, and then judge its efficiency by the extent to which that S.D. is altered by selection of the other characters.

(19) Professor Pearson has pointed out to me that the ideal index characters would be given if we calculated the seven directions of uncorrelated variables, that is, the principal axes of the correlation "ellipsoid." Thus, given $x_1, x_2, \dots x_7$ correlated variables, the seven uncorrelated variables are:

$$\begin{aligned} X_1 &= l_{11}x_1 + l_{12}x_2 + \dots + l_{17}x_7 \\ X_2 &= l_{21}x_1 + l_{22}x_2 + \dots + l_{27}x_7 \\ &\quad \&c. \qquad \&c. \end{aligned}$$

where the l 's give the directions of the principal axes, and $X_1, X_2, \dots X_7$ would be the proper index functions to identify criminals by if we have nothing better than the present correlated characters to work with. Of course this necessitates a preliminary determination of 49 numerical multipliers, but if these were once calculated, the uncorrelated characters of a criminal could be easily found from the measured correlated characters, and his identity established from an X -cabinet, into which we might enter *in any order*. I propose to return in a later paper to this calculation.

(20) *Reconstruction of Height from a knowledge of Finger, Cubit, and Foot, for medico-legal purposes.*

I now propose to consider in greater detail the four characters which have very high correlation, viz., Finger, Cubit, Foot and Height. For these four,

$$\Delta = \begin{vmatrix} 1 & .84638 & .75871 & .66084 \\ .84638 & 1 & .79699 & .79986 \\ .75871 & .79699 & 1 & .73636 \\ .66084 & .79986 & .73636 & 1 \end{vmatrix} = .031586,$$

and

$$\begin{aligned}
 R_{11} &= \cdot 121635, & \sqrt{\Delta/R_{11}} &= \cdot 509587, & R_{12} &= \cdot 088885 \\
 R_{22} &= \cdot 183825, & \sqrt{\Delta/R_{22}} &= \cdot 414520, & R_{13} &= -\cdot 031910 \\
 R_{33} &= \cdot 101914, & \sqrt{\Delta/R_{33}} &= \cdot 556710, & R_{14} &= -\cdot 014211 \\
 R_{44} &= \cdot 096392, & \sqrt{\Delta/R_{44}} &= \cdot 572436, & R_{23} &= \cdot 030695 \\
 & & & & R_{24} &= -\cdot 065693 \\
 & & & & R_{34} &= \cdot 029405
 \end{aligned}$$

Using the values of S.D. and Means given in Table 18 we can now reconstruct the various characters if one or more are given us; we shall choose Height (H) as being probably the most useful example.

(i) *Reconstruction of Height from Finger.*

$$\begin{aligned}
 H &= 166\cdot45716 + \frac{6\cdot45397}{\cdot 54786} \times \cdot 66084 \text{ (Finger - } 11\cdot54737\text{)}, \\
 &= 166\cdot45716 + 7\cdot7849 \text{ (Finger - } 11\cdot54737\text{)}.
 \end{aligned}$$

$$\begin{aligned}
 \text{Probable error} &= \cdot 67449 \times 6\cdot45397 \sqrt{1 - (\cdot 66084)^2/p}, \\
 &= 3\cdot2671/\sqrt{p}.
 \end{aligned}$$

(ii) *Reconstruction of Height from Cubit.*

$$H = 166\cdot45716 + 2\cdot6301 \text{ (Cubit - } 45\cdot05864\text{)}.$$

$$\text{Probable error} = 2\cdot6127/\sqrt{p}.$$

(iii) *Reconstruction of Height from Foot.*

$$H = 166\cdot45716 + 4\cdot0301 \text{ (Foot - } 25\cdot68770\text{)}.$$

$$\text{Probable error} = 2\cdot9453/\sqrt{p}.$$

(iv) *Reconstruction of Height from Finger and Cubit.*

In this case,

$$\begin{aligned}
 \Delta &= \begin{vmatrix} 1 & \cdot 66084 & \cdot 79986 \\ \cdot 66084 & 1 & \cdot 84638 \\ \cdot 79986 & \cdot 84638 & 1 \end{vmatrix} = \cdot 101914, \\
 R_{11} &= \cdot 28364, & R_{12} &= \cdot 01614, & R_{13} &= -\cdot 24054, \\
 \sigma_2 &= \cdot 54786, & & & \sigma_3 &= 1\cdot96275, \\
 m_2 &= 11\cdot54737, & & & m_3 &= 45\cdot05864.
 \end{aligned}$$

Therefore equation for H is:

$$H = 166\cdot45716 - \cdot 6703 \text{ (Finger - } 11\cdot54737\text{)} + 2\cdot7886 \text{ (Cubit - } 45\cdot05864\text{)}^*$$

$$\begin{aligned}
 \text{and probable error} &= \frac{\cdot 67449 \times 6\cdot45397}{\sqrt{p}} \sqrt{\frac{\cdot 101914}{\cdot 28364}}, \\
 &= 2\cdot6094/\sqrt{p}.
 \end{aligned}$$

* This equation is of special interest, it shows that for persons of a given cubit the longer the finger, the less the height.

(v) *Reconstruction of Height from Finger and Foot.*

In this case,

$$\Delta = \begin{vmatrix} 1 & \cdot 66084 & \cdot 73636 \\ \cdot 66084 & 1 & \cdot 75871 \\ \cdot 73636 & \cdot 75871 & 1 \end{vmatrix} = \cdot 183825,$$

$$R_{11} = \cdot 42436, \quad R_{12} = -\cdot 10216, \quad R_{13} = -\cdot 23497,$$

$$\sigma_2 = \cdot 54786, \quad \sigma_3 = 1\cdot 17923,$$

$$m_2 = 11\cdot 54737, \quad m_3 = 25\cdot 68770.$$

Therefore the equation becomes :

$$H = 166\cdot 45716 + 2\cdot 8360 (\text{Finger} - 11\cdot 54737) + 3\cdot 0304 (\text{Foot} - 25\cdot 68770),$$

and probable error $\quad \quad \quad = 2\cdot 8651/\sqrt{p}.$

(vi) *Reconstruction of Height from Cubit and Foot.*

In this case,

$$\Delta = \begin{vmatrix} 1 & \cdot 79986 & \cdot 73636 \\ \cdot 79986 & 1 & \cdot 79699 \\ \cdot 73636 & \cdot 79699 & 1 \end{vmatrix} = \cdot 121635,$$

$$R_{11} = \cdot 36481, \quad R_{12} = -\cdot 21299, \quad R_{13} = -\cdot 09888,$$

$$\sigma_2 = 1\cdot 96275, \quad \sigma_3 = 1\cdot 17923,$$

$$m_2 = 45\cdot 05864, \quad m_3 = 25\cdot 68770.$$

Therefore the equation is :

$$H = 166\cdot 45716 + 1\cdot 9198 (\text{Cubit} - 45\cdot 05864) + 1\cdot 4834 (\text{Foot} - 25\cdot 6877),$$

and probable error $\quad \quad \quad = 2\cdot 5136/\sqrt{p}.$

(vii) *Reconstruction of Height from Finger, Cubit and Foot.*

The Δ and R 's are already calculated, and from these we can form the equation in H . We will however calculate only the probable error, which is

$$\frac{\cdot 67449 \times 6\cdot 45397}{\sqrt{p}} \sqrt{\frac{\cdot 031586}{\cdot 096392}}$$

$$= 2\cdot 4919/\sqrt{p}.$$

(viii) *Reconstruction of Height from all the other six organs* (see formula in § 17).

This formula has not been worked out numerically, but its probable error

$$= \frac{\cdot 67449 \times 6\cdot 45397}{\sqrt{p}} \sqrt{\frac{\cdot 012129}{\cdot 038347}}$$

$$= 2\cdot 4482/\sqrt{p}.$$

We will now collect the probable errors which we have just calculated, taking $p = 1$, which is the case that most interests us.

TABLE 27.
Reconstruction of Height.

From	Probable Error
Finger	3·2671
Cubit	2·6127
Foot	2·9453
Finger and Cubit	2·6094
Finger and Foot	2·8651
Cubit and Foot	2·5136
Finger, Cubit and Foot	2·4919
Finger, Cubit, Foot, Head Length, Head Breadth and Face Breadth	2·4482

It will be noted that we do not improve much on the result obtained from Cubit and Foot by taking into account Finger also; nor on the result obtained from Cubit alone by taking into account Finger and Foot or even by the addition of all the remaining six characters.

As a practical problem, it is conceivable that a medico-legal case might arise where it would be important to discover the most probable height, given a foot or a cubit; the equations in (iii) and (ii) would enable us to reconstruct the height with a probable error of 2·9453 mm. and 2·6127 mm. respectively.

(21) *Examples of Reconstruction Formulae.* I will illustrate the sort of errors that will be made by the use of these formulæ by applying them to the reconstruction of stature in ten cases selected at random from my material.

The Table 28 shows the difference between the actual and the calculated heights, *minus* denoting that the calculated stature is in defect, *plus* that it is in excess.

It will be noticed that the result of calculating from Finger and Cubit is practically the same as from Cubit alone.

As was to be expected, the reconstruction is only approximate, but it is quite as good as the results obtained for the reconstruction of stature from the long bones*. Given the Cubit, the Finger and Cubit or the Foot and Cubit, we can estimate the stature with a probable error of about one inch. We see that the actual ten cases give deviations, plus and minus, clustered round the probable error of each method of estimating in quite a reasonable manner. Occasionally our estimate might be wrong by upwards of two inches, but such cases are rare. It is of course the old story: No formula will give the stature with very great closeness from the measurement of one or two other organs. If it did, organic relationship would be perfect correlation, variation would have ceased, and with it of necessity the Darwinian theory of evolution. Allowing accordingly for this limitation, I believe the results obtained to be of use for purposes of criminal investigation. For example a finger, cubit or foot having been discovered, and

* Pearson: On the Reconstruction of the Stature of Prehistoric Races, *Phil. Trans.* Vol. 192, A., p. 188.

a crime being suspected, the police might from my formulæ give a very fair estimate of the stature of the supposed victim for the purpose of identification. The form of such an estimate from the cubit, for my first individual, for example, would be $62'' \pm 1''$.

TABLE 28.

*Showing the difference in cms. between Actual and Calculated Height in
10 Criminals selected at random.*

Actual Stature, cms.	STATURE CALCULATED FROM						Average
	Finger	Cubit	Foot	Finger and Cubit	Finger and Foot	Foot and Cubit	
160.3	+ .3	- 2.4	+ .6	- 2.4	- .2	- 2.1	- 1.03
168.3	- 1.4	- 2.3	- 6.2	- 2.3	- 5.	- 3.8	- 3.50
170.2	- 2.6	- 4.2	- 3.3	- 4.3	- 3.	- 3.9	- 3.55
158.4	+ 5.4	+ 1.3	- 2.	+ 1.2	- .5	- .5	+ .82
161.9	+ 1.1	+ 3.1	+ 2.6	+ 3.3	+ 1.8	+ 2.8	+ 2.45
165.7	+ .3	- .4	+ 3.2	- .5	+ 2.5	+ .8	+ .98
167.6	+ 1.6	+ .5	+ 5.8	+ .4	+ 5.	+ 2.6	+ 2.65
160.	+ 4.5	+ 1.	+ 3.7	+ .9	+ 3.7	+ 1.5	+ 2.55
161.9	+ 8.1	+ 6.2	- .6	+ 6.	+ 1.9	+ 3.9	+ 4.25
169.5	- 5.	- 1.1	- 4.2	- .8	- 4.6	- 2.	- 2.95
Average	3.03	2.25	3.22	2.21	2.82	2.39	2.47
Prob. Error	± 3.3	± 2.6	± 2.9	± 2.6	± 2.9	± 2.5	± 2.8

(22) *Summary.* I will conclude with a short summary of the results set forth in detail in this paper.

(i) The great value of the new method of calculating standard deviations, means, and coefficients of correlation is demonstrated; it economises labour, and leads to results closely approximating to those obtained by more laborious methods.

(ii) Then as regards the criminals; they are shown to be homogeneous, fairly normal in the distribution of the selected characters, markedly different from the educated classes in stature, and in size and shape of the head, but in variability they agree generally with other classes and races of mankind.

(iii) Finally as regards the problem of identification, it has been shown that certain characters at present in use are not very suitable for the purpose, owing to their high correlation, but assuming them to be employed, we have discovered the proper order in which they should be entered in an index cabinet—an order very different from that determined by Scotland Yard—and have indicated a method of calculating uncorrelated characters which would furnish an ideal system of identification.

TABLE I. 3000 *Criminals*.
Head Length (centimetres).

[illegible]

TABLE II. 3000 Criminals. Height (feet and inches).

[illegible]

Head Breadth (millimetres).

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TABLE V. 1306 Criminals.

Height (feet and inches).

Head Breadth (millimetres).		Height (feet and inches).		Totals	
13-4	1	5-6	1	1	1
5	1	6	1	1	1
6	1	7	1	1	1
7	1	8	1	1	1
8	1	9	1	1	1
9	1	10	1	1	1
10	1	11	1	1	1
11	1	12	1	1	1
12	1	13	1	1	1
13	1	14	1	1	1
14	1	15	1	1	1
15	1	16	1	1	1
16	1	17	1	1	1
17	1	18	1	1	1
18	1	19	1	1	1
19	1	20	1	1	1
20	1	21	1	1	1
21	1	22	1	1	1
22	1	23	1	1	1
23	1	24	1	1	1
24	1	25	1	1	1
25	1	26	1	1	1
26	1	27	1	1	1
27	1	28	1	1	1
28	1	29	1	1	1
29	1	30	1	1	1
30	1	31	1	1	1
31	1	32	1	1	1
32	1	33	1	1	1
33	1	34	1	1	1
34	1	35	1	1	1
35	1	36	1	1	1
36	1	37	1	1	1
37	1	38	1	1	1
38	1	39	1	1	1
39	1	40	1	1	1
40	1	41	1	1	1
41	1	42	1	1	1
42	1	43	1	1	1
43	1	44	1	1	1
44	1	45	1	1	1
45	1	46	1	1	1
46	1	47	1	1	1
47	1	48	1	1	1
48	1	49	1	1	1
49	1	50	1	1	1
50	1	51	1	1	1
51	1	52	1	1	1
52	1	53	1	1	1
53	1	54	1	1	1
54	1	55	1	1	1
55	1	56	1	1	1
56	1	57	1	1	1
57	1	58	1	1	1
58	1	59	1	1	1
59	1	60	1	1	1
60	1	61	1	1	1
61	1	62	1	1	1
62	1	63	1	1	1
63	1	64	1	1	1
64	1	65	1	1	1
65	1	66	1	1	1
66	1	67	1	1	1
67	1	68	1	1	1
68	1	69	1	1	1
69	1	70	1	1	1
70	1	71	1	1	1
71	1	72	1	1	1
72	1	73	1	1	1
73	1	74	1	1	1
74	1	75	1	1	1
75	1	76	1	1	1
76	1	77	1	1	1
77	1	78	1	1	1
78	1	79	1	1	1
79	1	80	1	1	1
80	1	81	1	1	1
81	1	82	1	1	1
82	1	83	1	1	1
83	1	84	1	1	1
84	1	85	1	1	1
85	1	86	1	1	1
86	1	87	1	1	1
87	1	88	1	1	1
88	1	89	1	1	1
89	1	90	1	1	1
90	1	91	1	1	1
91	1	92	1	1	1
92	1	93	1	1	1
93	1	94	1	1	1
94	1	95	1	1	1
95	1	96	1	1	1
96	1	97	1	1	1
97	1	98	1	1	1
98	1	99	1	1	1
99	1	100	1	1	1
100	1	101	1	1	1
101	1	102	1	1	1
102	1	103	1	1	1
103	1	104	1	1	1
104	1	105	1	1	1
105	1	106	1	1	1
106	1	107	1	1	1
107	1	108	1	1	1
108	1	109	1	1	1
109	1	110	1	1	1
110	1	111	1	1	1
111	1	112	1	1	1
112	1	113	1	1	1
113	1	114	1	1	1
114	1	115	1	1	1
115	1	116	1	1	1
116	1	117	1	1	1
117	1	118	1	1	1
118	1	119	1	1	1
119	1	120	1	1	1
120	1	121	1	1	1
121	1	122	1	1	1
122	1	123	1	1	1
123	1	124	1	1	1
124	1	125	1	1	1
125	1	126	1	1	1
126	1	127	1	1	1
127	1	128	1	1	1
128	1	129	1	1	1
129	1	130	1	1	1
130	1	131	1	1	1
131	1	132	1	1	1
132	1	133	1	1	1
133	1	134	1	1	1
134	1	135	1	1	1
135	1	136	1	1	1
136	1	137	1	1	1
137	1	138	1	1	1
138	1	139	1	1	1
139	1	140	1	1	1
140	1	141	1	1	1
141	1	142	1	1	1
142	1	143	1	1	1
143	1	144	1	1	1
144	1	145	1	1	1
145	1	146	1	1	1
146	1	147	1	1	1
147	1	148	1	1	1
148	1	149	1	1	1
149	1	150	1	1	1
150	1	151	1	1	1
151	1	152	1	1	1
152	1	153	1	1	1
153	1	154	1	1	1
154	1	155	1	1	1
155	1	156	1	1	1
156	1	157	1	1	1
157	1	158	1	1	1
158	1	159	1	1	1
159	1	160	1	1	1
160	1	161	1	1	1
161	1	162	1	1	1
162	1	163	1	1	1
163	1	164	1	1	1
164	1	165	1	1	1
165	1	166	1	1	1
166	1	167	1	1	1
167	1	168	1	1	1
168	1	169	1	1	1
169	1	170	1	1	1
170	1	171	1	1	1
171	1	172	1	1	1
172	1	173	1	1	1
173	1	174	1	1	1
174	1	175	1	1	1
175	1	176	1	1	1
176	1	177	1	1	1
177	1	178	1	1	1
178	1	179	1	1	1
179	1	180	1	1	1
180	1	181	1	1	1
181	1	182	1	1	1
182	1	183	1	1	1
183	1	184	1	1	1
184	1	185	1	1	1
185	1	186	1	1	1
186	1	187	1	1	1
187	1	188	1	1	1
188	1	189	1	1	1
189	1	190	1	1	1
190	1	191	1	1	1
191	1	192	1	1	1
192	1	193	1	1	1
193	1	194	1	1	1
194	1	195	1	1	1
195	1	196	1	1	1
196	1	197	1	1	1
197	1	198	1	1	1
198	1	199	1	1	1
199	1	200	1	1	1
200	1	201	1	1	1
201	1	202	1	1	1
202	1	203	1	1	1
203	1	204	1	1	1
204	1	205	1	1	1
205	1	206	1	1	1
206	1	207	1	1	1
207	1	208	1	1	1
208	1	209	1	1	1
209	1	210	1	1	1
210	1	211	1	1	1
211	1	212	1	1	1
212	1	213	1	1	1
213	1	214	1	1	1
214	1	215	1	1	1
215	1	216	1	1	1
216	1	217	1	1	1
217	1	218	1	1	1
218	1	219	1	1	1
219	1	220	1	1	1
220	1	221	1	1	1
221	1	222	1	1	1
222	1	223	1	1	1
223	1	224	1	1	1
224	1	225	1	1	1
225	1	226	1	1	1
226	1	227	1	1	1
227	1	228	1	1	1
228	1	229	1	1	1
229	1	230	1	1	1
230	1	231	1	1	1
231	1	232	1	1	1
232	1	233	1	1	1
233	1	234	1	1	1
234	1	235	1	1	1
235	1	236	1	1	1
236	1	237	1	1	1
237	1	238	1	1	1
238	1	239	1	1	1
239	1	240	1	1	1
240	1	241	1	1	1
241	1	242	1	1	1
242	1	243	1	1	1
243	1	244	1	1	1
244	1	245	1	1	1
245	1	246	1	1	1
246	1	247	1	1	1
247	1	248	1	1	1
248	1	249	1	1	1
249	1	250	1	1	1
250	1	251	1	1	1
251	1	252	1	1	1
252	1	253	1	1	1
253	1	254	1	1	1
254	1	255	1	1	1
255	1	256	1	1	1
256	1	257	1	1	1
257	1	258	1	1	1
258	1	259	1	1	1
259	1	260	1	1	1
260	1	261	1	1	1
261	1	262	1	1	1
262	1	263	1	1	1
263	1	264	1	1	1
264	1	265	1	1	1
265	1	266	1	1	1
266	1	267	1	1	1
267	1	268	1	1	1
268	1	269	1	1	1
269	1	270	1	1	1
270	1	271	1	1	1
271	1	272	1	1	1
272	1	273	1	1	1
273	1	274	1	1	1
274	1	275	1	1	1
275	1	276	1	1	1
276	1	277	1	1	1
277	1	278	1	1	1
278	1	279	1	1	1
279	1	280	1	1	1
280	1	281	1	1	1
281	1	282	1	1	1
282	1	283	1	1	1
283	1	284	1	1	

TABLE VI. 1306 *Criminals*. Height (feet and inches).

Totals	0	1	0	0	1	1	4	1	5	10	10	13	21	32	33	51	74	60	88	73	94	104	98	104	74	77	79	44	28	41	22	21	12	12	7	3	4	0	3	0	0	1	1306		
$\frac{91}{6}, \frac{9}{6} - \frac{91}{6}, \frac{7}{6}$																			1																								1		
$\frac{91}{6}, \frac{7}{6} - \frac{91}{6}, \frac{8}{6}$																																													
$\frac{91}{6}, \frac{8}{6} - \frac{91}{6}, \frac{8}{6}$																																													
$\frac{91}{6}, \frac{8}{6} - \frac{91}{6}, \frac{7}{6}$																															1														
$\frac{91}{6}, \frac{7}{6} - \frac{91}{6}, \frac{0}{6}$																					1												1			1					1	4			
$\frac{91}{6}, \frac{0}{6} - \frac{91}{6}, \frac{11}{6}, \frac{9}{6}$																																	1			1	1					3			
$\frac{91}{6}, \frac{11}{6} - \frac{91}{6}, \frac{0}{6}, \frac{10}{6}$																						2	1	1	1	1	1	1	2	1	3	3	5	2	2	2	2	1			2			26	
$\frac{91}{6}, \frac{0}{6}, \frac{10}{6} - \frac{91}{6}, \frac{6}{6}$																					1																							37	
$\frac{91}{6}, \frac{6}{6} - \frac{91}{6}, \frac{8}{6}$																		1	2	3	5	6	8	10	6	5	5	4	6	3	3	2	2	4	2	2	3								80
$\frac{91}{6}, \frac{8}{6} - \frac{91}{6}, \frac{11}{6}$																			3	3	7	13	7	10	6	5	8	10	11	6	3	8	2	2	4	1	3	1							103
$\frac{91}{6}, \frac{11}{6} - \frac{91}{6}, \frac{9}{6}$												1										10	13	14	15	17	17	10	8	3	5	8	5	1	1	4	1	1			1			175	
$\frac{91}{6}, \frac{9}{6} - \frac{91}{6}, \frac{2}{6}$													1	1	6	6	7					21	17	14	14	17	9	4	4	3	8	2	2	4	1									193	
$\frac{91}{6}, \frac{2}{6} - \frac{91}{6}, \frac{7}{6}$								1					1	2	9	12	10					20	21	18	14	14	9	3	4	6	2														193
$\frac{91}{6}, \frac{7}{6} - \frac{91}{6}, \frac{8}{6}$										1	1	3	3	6	3	12	16	12	13	15	9	10	10	14	15	17	17	10	8	3	4														171
$\frac{91}{6}, \frac{8}{6} - \frac{91}{6}, \frac{8}{6}$								1	3	2	2	9	3	7	16	12	12	8	11	10	13	12	17	12	18	20	16	16	5	5	6	6	3	4		1									151
$\frac{91}{6}, \frac{8}{6} - \frac{91}{6}, \frac{7}{6}$					1	1	2		1	2	1	3	7	7	5	11	10	9	2	12	5	7	12	7	12	17	12	4	1	1															81
$\frac{91}{6}, \frac{7}{6} - \frac{91}{6}, \frac{0}{6}$		1							1	2	2	4		4	3	5	5	7	4	3	2	2		2	2		1																		48
$\frac{91}{6}, \frac{0}{6} - \frac{91}{6}, \frac{11}{6}, \frac{9}{6}$					1		1	1	1	1	1	2	3	3	2	2	4		1			2					1																	23	
$\frac{91}{6}, \frac{11}{6} - \frac{91}{6}, \frac{0}{6}, \frac{10}{6}$					1	1	1	1	2	1	1	1	2	2		1	1	1																										15	
$\frac{91}{6}, \frac{0}{6}, \frac{10}{6} - \frac{91}{6}, \frac{6}{6}$																																													
$\frac{91}{6}, \frac{6}{6} - \frac{91}{6}, \frac{8}{6}$							1																																					1	
$\frac{91}{6}, \frac{8}{6} - \frac{91}{6}, \frac{11}{6}$																																													
	9-4	5	6	7	8	9	10-0	1	2	3	4	5	6	7	8	9	10-0	1	2	3	4	5	6	7	8	9	10-0	1	2	3	4	5	6	7	8	9	10-0	1	2	3	4	5	Totals		

Left Middle Finger (millimetres).

TABLE VII.

1000 Cambridge Men.

Head Length (ins.)

Head Breadth (ins.)	6·8	6·9	7	7·1	7·2	7·3	7·4	7·5	7·6	7·7	7·8	7·9	8	8·1	8·2	8·3	8·4	8·5	8·6	Totals
5·5	—	—	—	—	1	—	2	—	—	—	—	—	—	—	—	—	—	—	—	3
5·6	—	—	—	—	2	3	1	4	2	—	—	—	—	—	—	—	—	—	—	12
5·7	1	1	—	1	2	6	5	7	5	8	3	4	—	—	—	—	—	—	—	43
5·8	—	—	2	4	6	7	15	12	12	11	7	3	1	—	—	—	—	—	—	80
5·9	—	—	1	4	6	16	16	24	20	28	7	5	4	—	—	—	—	—	—	131
6	—	1	—	6	14	12	24	40	45	40	28	9	13	4	—	—	—	—	—	236
6·1	—	—	1	3	4	6	11	27	42	40	22	15	10	2	2	—	—	—	—	185
6·2	—	—	1	2	6	3	14	18	22	25	13	14	17	2	4	1	—	—	—	142
6·3	—	—	1	—	1	4	9	8	16	15	15	16	6	7	—	—	—	1	—	99
6·4	—	—	—	—	1	—	2	4	6	6	3	4	7	2	1	1	—	—	—	37
6·5	—	—	—	—	—	1	1	—	3	3	—	2	3	1	1	—	—	—	—	15
6·6	—	—	—	1	—	—	—	1	2	3	—	1	—	2	1	1	—	—	—	12
6·7	—	—	—	—	—	—	—	—	—	1	—	—	—	2	—	—	—	—	—	3
6·8	—	—	—	—	—	—	—	—	—	—	—	2	—	—	—	—	—	—	—	2
Totals	1	2	6	21	43	58	100	145	175	180	98	75	61	22	9	3	—	—	1	1000

TABLE VIII.

1000 Cambridge Men.

Height (feet and inches).

Head Breadth (ins.)	5' 1½-2½	2½-3½	3½-4½	4½-5½	5½-6½	6½-7½	7½-8½	8½-9½	9½-10½	10½-11½	11½-6'0½	6' 0½-1½	1½-2½	2½-3½	3½-4½	4½-5½	Totals
5·5	—	—	—	—	—	1	1	—	—	0·5	0·5	—	—	—	—	—	3
5·6	0·5	0·5	—	1	1	3	1·5	1·5	0·5	1	1·5	—	—	—	—	—	12
5·7	1	—	—	—	2	5·5	6·5	9	10	5	2·	1	—	—	—	—	43
5·8	—	3	4	6	11	13·5	11	12	9·5	5·5	3·5	—	—	—	1	—	80
5·9	2	4	3·5	4·5	14	16	19	20·5	22	8·5	12·5	2	1·5	1	—	—	131
6	—	4	6	5·5	22·5	32·5	42·5	38	26	28·5	8·5	14	6	1	—	—	236
6·1	—	3	5	9	12	27	22·5	35	24·5	23·5	8·5	4·5	8	2·5	—	—	185
6·2	0·5	3·5	4	6·5	11·5	12·5	15·5	28	24·5	16	5·5	7	2	3	1·5	0·5	142
6·3	—	—	2	7	5	11·5	9·5	22	12·5	13·5	4·5	8·5	—	2·5	0·5	—	99
6·4	—	1	—	—	3	0·5	5	8	4	4	4·5	6	1	—	—	—	37
6·5	—	—	—	—	—	—	1	4	3	1	1	2·5	0·5	2	—	—	15
6·6	—	—	—	1	1·5	0·5	4	1	—	1	1	2	—	—	—	—	12
6·7	—	—	—	—	1	—	—	—	—	—	—	—	2	—	—	—	3
6·8	—	—	—	—	—	—	—	—	2	—	—	—	—	—	—	—	2
Totals	4	19	24·5	40·5	84·5	123·5	139	179	138·5	108	53·5	47·5	21	12	5	0·5	1000

TABLE IX.
1000 Cambridge Men.
Head Length (inches).

Height (feet and inches).		6-8	6-9	7	7-1	7-2	7-3	7-4	7-5	7-6	7-7	7-8	7-9	8	8-1	8-2	8-3	8-4	8-5	8-6	Totals
5'	11-2 1/2	—	1	—	—	—	—	—	2	0-5	—	—	—	—	—	—	—	—	—	—	4
	11-3 1/2	—	—	1	2	2-5	2	—	4-5	2-	2	—	—	—	—	—	—	—	—	—	19
	11-4 1/2	—	—	—	1	2	—	5-5	6	5-5	4-5	—	—	—	—	—	—	—	—	—	24-5
	11-5 1/2	—	—	—	—	3	4	3	6-5	4-5	4-5	—	—	—	—	—	—	—	—	—	40-5
	11-6 1/2	1	—	—	5-5	1	6	10-5	14	21	11	7	7	2	1	—	—	—	—	—	84-5
	11-7 1/2	—	—	—	1-5	8-5	10	14-5	20	21-5	19-5	12	7-5	6	1	0-5	—	—	—	—	123-5
	11-8 1/2	—	—	—	2-5	8-5	6-5	22-5	24-5	29	24	9-5	4	3	1	1-5	—	—	—	—	139
	11-9 1/2	—	—	—	5-5	4	8-5	16-5	24-5	31-5	36-5	14	21-5	12	3	1	—	—	—	—	179
	11-10 1/2	—	—	—	2	4	8-5	15	13-5	21-5	26	14-5	13-5	13	4	1-5	1	—	—	—	138-5
	11-11 1/2	—	—	—	—	3	7-5	3-5	12	22-5	25-5	16	10	6-5	1	0-5	—	—	—	—	108
6'	11-6 0 1/2	—	—	—	—	3	4	5	6-5	8-5	9	5-5	3	5	3	1	—	—	—	—	53-5
	11-7 0 1/2	—	—	—	—	2	1	2	6	2	8-5	7-5	5	5	3	1-5	2	—	—	1	47-5
	11-8 0 1/2	—	—	—	—	—	—	—	3	2	4-5	2-5	1	5-5	2	0-5	—	—	—	—	21
	11-9 0 1/2	—	—	—	—	1	—	—	—	2	4-5	1	—	3	0-5	—	—	—	—	—	12
	11-10 0 1/2	—	—	—	—	—	—	—	—	1	—	1	—	—	—	—	—	—	—	—	5
	11-11 0 1/2	—	—	—	—	—	—	—	0-5	—	—	—	—	—	—	—	—	—	—	—	0-5
	11-12 0 1/2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Totals		1	2	6	21	43	58	100	145	175	180	98	75	61	22	9	3	—	—	1	1000

TABLE XI.

Height (ins.)

Head Breadth (cms.)	under $64\frac{9}{16}$	$64\frac{9}{16}$ to $66\frac{9}{16}$	over $66\frac{9}{16}$	Totals
14.8 and under	455	323	299	1077
14.9 to 15.2 ...	316	277	313	906
over 15.2 ...	283	320	414	1017
Totals	1054	920	1026	3000

TABLE XII.

Height (ins.)

L. M. Finger (cms.)	under $64\frac{9}{16}$	over $64\frac{9}{16}$	Totals
11.5 and under	849	665	1514
over 11.5 ...	205	1281	1486
Totals	1054	1946	3000

TABLE XIII.

Height (ins.)

L. M. Finger (cms.)	under $64\frac{9}{16}$	$64\frac{9}{16}$ to $66\frac{9}{16}$	over $66\frac{9}{16}$	Totals
11.1 and under	515	149	42	706
11.2 to 11.9 ...	504	607	501	1612
over 11.9 ...	35	164	483	682
Totals	1054	920	1026	3000

TABLE XIV.

Head Length (cms.)

Head Breadth (cms.)	19.1 and under	over 19.1	Totals
14.8 and under	705	372	1077
over 14.8 ...	750	1173	1923
Totals	1455	1545	3000

TABLE XV.

Head Length (cms.)

Head Breadth (cms.)	18.5 and under	18.6 to 19.7	over 19.7	Totals
14.8 and under	279	728	70	1077
14.9 to 15.2 ...	113	670	123	906
over 15.2 ...	71	645	301	1017
Totals	463	2043	494	3000

TABLE XVI.
Head Length (cms.)

Face Breadth (cms.)		19.1 and under	over 19.1	Totals
	13.6 and under	921	579	1500
	over 13.6 ...	534	966	1500
	Totals	1455	1545	3000

TABLE XVII.
Head Length (cms.)

Face Breadth (cms.)		18.8 and under	18.9 to 19.3	over 19.3	Totals
	13.4 and under	453	357	229	1039
	13.5 to 13.8 ...	259	335	328	922
	over 13.8 ...	146	333	560	1039
	Totals	858	1025	1117	3000

TABLE XVIII.
Head Length (cms.)

Left Cubit (cms.)		19.1 and under	over 19.1	Totals
	44.9 and under	836	583	1419
	over 44.9 ...	619	962	1581
	Totals	1455	1545	3000

TABLE XIX.
Head Length (cms.)

Left Cubit (cms.)		18.8 and under	18.9 to 19.3	over 19.3	Totals
	44 and under	375	310	226	911
	44.1 to 45.7	264	365	373	1002
	over 45.7 ...	219	350	518	1087
	Totals	858	1025	1117	3000

TABLE XX.
Head Length (cms.)

Left Foot (cms.)		19.1 and under	over 19.1	Totals
	25.5 and under	832	545	1377
	over 25.5 ...	623	1000	1623
	Totals	1455	1545	3000

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TABLE XXI.

Head Length (cms.)

Left Foot (cms.)		18.8 and under	18.9 to 19.3	over 19.3	Totals
	25 and under	378	300	205	883
	25.1 to 26 ...	268	345	366	979
	over 26 ...	212	380	546	1138
	Totals	858	1025	1117	3000

TABLE XXII.

Head Length (cms.)

L. M. Finger (cms.)		19.1 and under	over 19.1	Totals
	11.5 and under	880	634	1514
	over 11.5 ...	575	911	1486
	Totals	1455	1545	3000

TABLE XXIII.

Height (ins.)

Head Length (cms.)		64 $\frac{9}{16}$ and under	over 64 $\frac{9}{16}$	Totals
	19.1 and under	665	790	1455
	over 19.1 ...	389	1156	1545
	Totals	1054	1946	3000

TABLE XXIV.

Head Breadth (cms.)

Face Breadth (cms.)		14.8 and under	over 14.8	Totals
	13.6 and under	834	666	1500
	over 13.6 ...	243	1257	1500
	Totals	1077	1923	3000

TABLE XXIV^a.

Head Breadth (cms.)

Face Breadth (cms.)		15 and under	over 15	Totals
	13.6 and under	1077	423	1500
	over 13.6 ...	451	1049	1500
	Totals	1528	1472	3000

TABLE XXV.
Head Breadth (cms.)

L. M. Finger (cms.)		14.8 and under	over 14.8	Totals
	11.5 and under	611	903	1514
	over 11.5 ...	466	1020	1486
	Totals	1077	1923	3000

TABLE XXV^a.
Head Breadth (cms.)

L. M. Finger (cms.)		15 and under	over 15	Totals
	11.5 and under	841	673	1514
	over 11.5 ...	687	799	1486
	Totals	1528	1472	3000

TABLE XXVI.
Head Breadth (cms.)

Left Cubit (cms.)		14.8 and under	over 14.8	Totals
	44.9 and under	570	849	1419
	over 44.9 ...	507	1074	1581
	Totals	1077	1923	3000

TABLE XXVII.
Left Foot (cms.)

Head Breadth (cms.)		25.5 and under	over 25.5	Totals
	14.8 and under	587	490	1077
	over 14.8 ...	790	1133	1923
	Totals	1377	1623	3000

TABLE XXVIII.
Face Breadth (cms.)

L. M. Finger (cms.)		13.6 and under	over 13.6	Totals
	11.5 and under	913	601	1514
	over 11.5 ...	587	899	1486
	Totals	1500	1500	3000

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TABLE XXIX.

Face Breadth (cms.)

Left Cubit (cms.)		13·6 and under	over 13·6	Totals
	44·9 and under	849	570	1419
	over 44·9 ...	651	930	1581
	Totals	1500	1500	3000

TABLE XXX.

Face Breadth (cms.)

Left Foot (cms.)		13·6 and under	over 13·6	Totals
	25·5 and under	865	512	1377
	over 25·5 ...	635	988	1623
	Totals	1500	1500	3000

TABLE XXXI.

Face Breadth (cms.)

Height (ins.)		13·6 and under	over 13·6	Totals
	64 $\frac{9}{16}$ and under	683	371	1054
	over 64 $\frac{9}{16}$...	817	1129	1946
	Totals	1500	1500	3000

TABLE XXXII.

Left Foot (cms.)

L. M. Finger (cms.)		25·5 and under	over 25·5	Totals
	11·5 and under	1103	411	1514
	over 11·5 ...	274	1212	1486
	Totals	1377	1623	3000

TABLE XXXIII.

L. M. Finger (cms.)

Left Cubit (cms.)		11·5 and under	over 11·5	Totals
	44·9 and under	1196	223	1419
	over 44·9 ...	318	1263	1581
	Totals	1514	1486	3000

TABLE XXXIV.

Left Foot (cms.)

Left Cubit (cms.)		25.5 and under	over 25.5	Totals
	44.9 and under	1089	330	1419
	over 44.9 ...	288	1293	1581
	Totals	1377	1623	3000

TABLE XXXV.

Height (ins.)

Left Cubit (cms.)		65 $\frac{9}{16}$ and under	over 65 $\frac{9}{16}$	Totals
	44.9 and under	1158	261	1419
	over 44.9 ...	358	1223	1581
	Totals	1516	1484	3000

TABLE XXXV^a.

Height (ins.)

Left Cubit (cms.)		64 $\frac{9}{16}$ and under	over 64 $\frac{9}{16}$	Totals
	44.9 and under	902	517	1419
	over 44.9 ...	152	1429	1581
	Totals	1054	1946	3000

TABLE XXXVI.

Height (ins.)

Left Foot (cms.)		65 $\frac{9}{16}$ and under	over 65 $\frac{9}{16}$	Totals
	25.5 and under	1088	289	1377
	over 25.5 ...	428	1195	1623
	Totals	1516	1484	3000

TABLE XXXVI^a.

Height (ins.)

Left Foot (cms.)		64 $\frac{9}{16}$ and under	over 64 $\frac{9}{16}$	Totals
	25.5 and under	873	504	1377
	over 25.5 ...	181	1442	1623
	Totals	1054	1946	3000